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Class incremental learning with KL constraint and multi-strategy exemplar selection for classification based on MMFA model

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ABSTRACT

Class incremental learning (CIL) can learn new classes continuously by updating the model rather than retraining a model from scratch with all seen classes like traditional offline learning, therefore, CIL is more suitable for classification in dynamic environments, where new classes are captured progressively. However, the key knowledge of old classes will be lost due to the update mode of CIL, leading to the catastrophic forgetting (CF) problem. In this paper, a novel CIL method with Kullback-Leibler constraint and multi-strategy exemplar selection (CIL-KLMES) is proposed for classification based on max-margin factor analysis (MMFA) model. To handle the CF problem, CIL-KLMES imposes a Kullback-Leibler (KL) divergence term on the important parameters when updating the model to restrict the parameters' distribution to be similar, thus preventing the updated model from deviating too much from the previous model and preserving the knowledge of old classes. Moreover, CIL-KLMES selects a few representative exemplars from old classes based on the robust description of data distribution and classification decision boundary. By replaying representative exemplars to update the model together with new class data, the key knowledge of old classes can be further preserved. Therefore, the CF problem can be alleviated sufficiently. Experimental results demonstrate the effectiveness of CIL-KLMES.

1. Introduction

Many classification methods have been proposed, where the probabilistic statistical models [1–9], including but not limited to Latent Dirichlet Allocation and Factor Analysis (FA) models, have been widely used in many areas due to their stronger generalization and the ability to incorporate prior knowledge. Especially, max-margin factor analysis (MMFA) model [6] jointly learns the probabilistic latent feature and max-margin classifier to yield more discriminative features in latent subspaces and maintain the strong data description ability, thus achieving excellent classification performance. Although achieving good classification performance, the above classification methods are in traditional offline learning mode, in which the model is usually trained with all samples of all seen classes. In real-world applications, data are dynamic where new classes might be captured progressively. For example, in satellite target recognition [10–12], new satellite targets enter the orbit continuously, and new classes are gradually obtained. When new classes arrive, the traditional offline learning mode has to retrain the model from scratch with both the past and new class data, leading to huge memory requirements. Due to memory limitations in practice, storing all data from the past classes may be not realistic, which limits the application of the traditional offline learning mode in dynamic environments. Class incremental learning

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(CIL), which can update the model with new class data and a few stored data from old classes or only with new class data, is an ideal learning mechanism.

Since no data or only a few data of old classes participate in the model update in CIL, the learned knowledge of old classes may be lost, resulting in severe performance degradation on old classes. This is known as the catastrophic forgetting (CF) [13–19] problem. Current CIL methods mainly focus on how to alleviate the CF problem to maintain the classification performance on old classes. Some existing CIL methods restrict the parameter update at the parameter level, like Elastic Weight Consolidation (EWC) [20], class incremental extreme learning machine (CIELM) [21] and Learning without Memorizing (LwM) [22]. EWC introduces an additional regularization term into loss function to restrict the update of important parameters. CIELM is proposed for ECG-based user identification, in which the model is updated after adding the new output nodes corresponding to new classes with some parameters frozen. LwM applies knowledge distillation to force the classification logits of old and new models to be similar, thus transferring the old knowledge from the old model into the new model. However, the preservation of the previous learned knowledge by restricting the parameter updates is insufficient, and the ability to alleviate the forgetting problem is limited [23–25]. For better alleviating the forgetting problem, some memory-replay-based CIL methods [15,19,25–30] are proposed. Memory-replay-based CIL methods mainly divide into two categories: pseudo instance rehearsal and exemplar rehearsal [31], i.e., generated pseudo instances or a few stored exemplars of old classes are replayed to update the model along with new class data in CIL phases. Fisher Discriminant Analysis Random Forest (FDARF) [27] is a CIL method based on pseudo instance rehearsal. In FDARF, the old knowledge is preserved by storing the statistics of data distribution, like the mean and the covariance matrix. When new classes arrive, pseudo instances are generated from the stored distribution to participate in the model update together with new class data. Recently, Sun et al. propose Discriminative and Comparable Parallel One-class Classifiers (DCPOC) [30]. The pseudo samples of old classes are generated via variational auto-encoders to update the model together with new classes. However, instead of directly storing representative exemplars, generating pseudo samples leads to more information loss. Some other memory-replay-based CIL methods preserve old knowledge by directly storing representative exemplars. This type of CIL methods focuses on how to select the exemplars with more representativeness to preserve the knowledge of old classes more comprehensively. Existing methods apply various exemplar selection methods. Herding [32] is one of the most common exemplar selection methods, which is first utilized in Incremental classifier and representation learning (iCaRL) [15]. By Herding, the exemplars whose features surround the mean feature of each class are selected and stored to participate in subsequent model updates. Besides iCaRL, some recent CIL methods [26,29] utilize Herding as the exemplar selection method. Later, Dang et al. recently develop class boundary exemplar selection-based incremental learning (CBesIL) [19] for radar target recognition. The exemplar selection method is based on local geometrical information, i.e., the location of data in the feature space, to select the exemplars from the interior and edge regions of each class and overlap region between classes according to three basic data distribution types defined in [33]. In [25], density coverage-based exemplar selection method (DCBes) is utilized to select the exemplars to achieve the maximum coverage of the data based on the metric learning theory and the set covering theory, by which the exemplars are mainly distributed in areas with higher density. However, since iCaRL, CBesIL and DCBes only consider local statistical information such as the mean and the local geometrical information, the integrality of preserved knowledge of old classes is weak, so that the ability to alleviate forgetting problems is limited.

Considering the strengths and shortcomings of the existing CIL methods for alleviating the CF problem, a novel CIL method with Kullback-Leibler constraint and multi-strategy exemplar selection (CIL-KLMES) based on a well-performed MMFA model is proposed for classification in dynamic environments. The method is built under the probabilistic statistical framework. In CIL-KLMES, the CF problem is alleviated from two aspects: parameter update constraint and multi-strategy exemplar selection. For parameter update constraint, CIL-KLMES imposes a Kullback-Leibler (KL) divergence term on the important parameters to restrict the distribution of these parameters updated at two adjacent incremental phases to be close, which can prevent the updated model from deviating too much from the previous model; hence, the previous learned knowledge of the past classes can be preserved at the parameter level. For further preserving the knowledge of old classes, we propose a more effective exemplar selection method to select the representative exemplars from the aspects of data distribution description and classification decision boundary preservation participating in the model update. In detail, we select a few samples with larger likelihood values and a few samples which are close to the classification decision boundary. The samples with larger likelihood values can better represent the data distribution, and the samples closer to the classification decision boundary have a greater influence on classification. Replaying these samples can help robustly describe the data distribution and the classification decision boundary of old classes. The exemplars selected based on multiple strategies are more representative, and replaying these exemplars can ensure the integrality of the knowledge preservation of old classes; thus, the learned knowledge of the past classes can be further preserved to alleviate the CF problem at the data level. To sum up, our proposed CIL-KLMES restricts the parameter update by KL constraint and selects the representative exemplars with more representativeness based on multiple strategies. Therefore, the CF problem can be more effectively alleviated.

Furthermore, the parameter estimation is performed via stochastic variational inference (SVI) algorithm [34]. By combining stochastic gradient descent (SGD) algorithm [35] with variational inference [36–38], SVI can update the model based on the existing model without retraining the model, making it suitable for CIL in the probabilistic-statistical framework. The experiments on several UCI datasets, simulated satellite high-resolution range profile (HRRP) dataset and Split CIFAR-10 dataset demonstrate the effectiveness of our CIL-KLMES method. Our main contributions are summarized as follows:

- (1) Our CIL-KLMES imposes a KL divergence term to prevent the updated model from deviating too much from the previous model, thus alleviating the CF problem at the parameter level.

- (2) A more effective multi-strategy exemplar selection method is developed based on the robust description of data distribution and classification decision boundary. Selecting the exemplars based on multiple strategies can ensure the integrality of the knowledge preservation of the past classes, thus further alleviating CF problem at the data level.
- (3) The parameters are effectively inferred via SVI algorithm, which combines SGD algorithm with variational inference. The parameter update mechanism of SVI makes it suitable for CIL in the probabilistic-statistical framework.

Table A.1 in Appendix lists the notations of the important variables in our CIL-KLMES to assist readers clearly understand the definitions of notations.

2. Preliminaries

2.1. Offline MMFA model

MMFA model utilizes FA model as the feature extractor and the latent variable support vector machine (LVSVM) [39] as the classification criterion in a united framework, of which the interplay between FA model and LVSVM can yield more discriminative features in latent subspaces while maintaining the strong data description ability. We start with a concise introduction of FA model, which serves as the core structure of MMFA model and describes the correlations among multidimensional data through a low-dimensional latent variable. Given a P -dimension training data set $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$ with N being the size of $\mathbf{X}$, the FA model assumes that $\mathbf{x}_i$ is generated by linearly mapping from a latent variable and adding an observation noise, which is expressed as follows:

$$\mathbf{x}_i = \mathbf{D}\mathbf{s}_i + \boldsymbol{\mu} + \boldsymbol{\epsilon}_i \quad (1)$$

$$\mathbf{d}_k \sim \mathcal{N}(0, \boldsymbol{\alpha}_k^{-1}), \alpha_{kp} \sim Ga(a_0, b_0) \quad (2)$$

$$s_{ki} \sim \mathcal{N}(0, \beta_k^{-1}), \beta_k \sim Ga(c_0, d_0) \quad (3)$$

$$\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \boldsymbol{\gamma}^{-1}), \gamma_p \sim Ga(e_0, f_0) \quad (4)$$

where $\boldsymbol{\mu}$ is the mean vector of $\mathbf{X}$; $\mathbf{d}_k$ is the k th column vector of the factor loading matrix $\mathbf{D} \in \mathbb{R}^{P \times K}$ with $\mathbf{d}_k$ obeying the Gaussian distribution as $\mathbf{d}_k \sim \mathcal{N}(0, \boldsymbol{\alpha}_k^{-1})$ and K being the number of factors ($P \gg K$); α_{kp} is the p th diagonal element of the precision $\boldsymbol{\alpha}_k$ obeying the Gamma distribution as $\alpha_{kp} \sim Ga(a_0, b_0)$. $\mathbf{s}_i \in \mathbb{R}^K$ is the latent variable with s_{ki} being the k th element of $\mathbf{s}_i$ and $s_{ki} \sim \mathcal{N}(0, \beta_k^{-1})$; $\boldsymbol{\epsilon}_i \in \mathbb{R}^P$ is the noise variable with $\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, \boldsymbol{\gamma}^{-1})$, and γ_p is the p th diagonal element of $\boldsymbol{\gamma}$. Similar to α_{kp} , the precisions β_k and γ_p obey Gamma distributions as $\beta_k \sim Ga(c_0, d_0)$ and $\gamma_p \sim Ga(e_0, f_0)$ respectively; $a_0, b_0, c_0, d_0, e_0, f_0$ are preset hyper-parameters. Besides the FA model, which is an unsupervised model, LVSVM is another important component of MMFA model to introduce label information, and the LVSVM structure in MMFA can be expressed as:

$$\mathbf{W} \sim \mathcal{N}(0, \sigma_{\mathbf{W}}^{-1} \mathbf{I}_{K+1}), \sigma_{\mathbf{W}} \sim Ga(\tau_1, \tau_2) \quad (5)$$

$$\phi(y_i, \lambda_i | \mathbf{W}) = \frac{1}{\sqrt{2\pi\lambda_i}} \exp\left(-\frac{(\lambda_i + h_0(1 - y_i \mathbf{W}^T \tilde{\mathbf{s}}_i))^2}{2\lambda_i}\right) \quad (6)$$

where the weighted coefficient $\mathbf{W}$ obeys the Gaussian distribution as $\mathbf{W} \sim \mathcal{N}(0, \sigma_{\mathbf{W}}^{-1} \mathbf{I}_{K+1})$ with $\mathbf{I}_{K+1}$ denoting $K+1$ dimensional identity matrix, and $\sigma_{\mathbf{W}}$ is the precision of $\mathbf{W}$ with τ_1 and τ_2 being the preset hyper-parameters. $\phi(y_i, \lambda_i | \mathbf{W})$ is a joint distribution, which is conjugate to multivariate normal prior distribution. h_0 is a positive tuning parameter, and λ_i is introduced as the augmented variable. $y_i \in \{-1, +1\}$ and $\tilde{\mathbf{s}}_i = [\mathbf{s}_i; 1]$ is the label and the augmented feature vector of $\mathbf{x}_i$ respectively. If $y_i = +1$, $\mathbf{x}_i$ is a positive sample, else being a negative sample. For further illustration, the graphical structure of MMFA model is shown in Fig. 1. The blue dotted box shows the FA model structure, which depicts the generation of sample $\mathbf{x}_i$; the red dotted box shows the LVSVM structure, which depicts the mapping process from the latent feature $\mathbf{s}_i$ of $\mathbf{x}_i$ to its label y_i . In Fig. 1, the square nodes are initialized hyperparameters and the circular nodes are variables, where $\mathbf{x}_i$ and y_i are observed variables, and the rest are unobserved variables. It is clear that the data generation and classification process of $\mathbf{x}_i$ are combined through $\mathbf{s}_i$. In other words, $\mathbf{s}_i$ is determined by not only the data $\mathbf{x}_i$ but also its corresponding label y_i , thus guaranteeing the linear separability of the data in the hidden space.

2.2. Class incremental learning

Recently, CIL for classification in dynamic environments has received much attention. In CIL, the model is updated based on the existing model with new class data and few stored data from old classes or only with new class data, like the manner of human cognition, instead of relearning a model with all data of all seen classes. To be specific, CIL assumes a scenario that data in streaming mode arrive over time, which can be represented as $\{\hat{\mathbf{X}}^0, \hat{\mathbf{X}}^1, \dots, \hat{\mathbf{X}}^t, \dots\}$. $\hat{\mathbf{X}}^t = \{\mathbf{X}_c^t\}_{c=1}^{C_t}$ denotes the training set obtained at time t with C_t being the number of classes in $\hat{\mathbf{X}}^t$ and $\mathbf{X}_c^t$ being the data of class c , where the classes of different training sets are mutually exclusive. The training process starts with $\hat{\mathbf{X}}^0$ at time 0, and the initial model $\mathcal{M}_0$ is trained with $\hat{\mathbf{X}}^0$ to classify C_0 initial classes. On the condition that no data of old classes are stored, the model is directly updated with new class data $\hat{\mathbf{X}}^t$ when new classes arrive at time t ($t \geq 1$). For memory-replay-based CIL methods, the flowchart is shown in Fig. 2. Fig. 2 depicts that the exemplar set $\mathbf{B}^{t-1}$

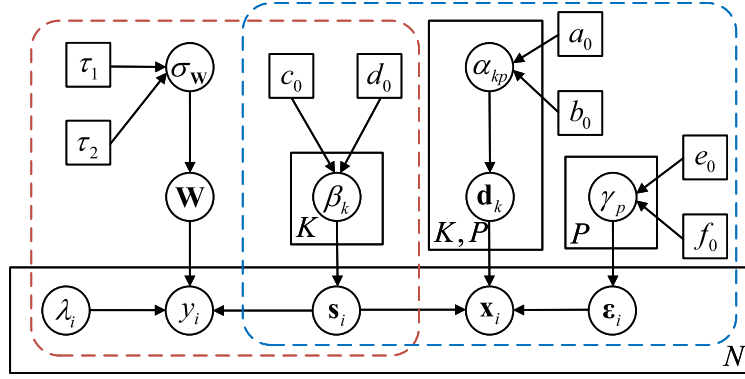


Fig. 1. Graphical structure of MMFA model. The symbol i is the sample index, and k is the latent factor index. The symbols N , K and P at the edges indicate a stack of boxes for indexes i , k and p , where N denotes the number of samples, K denotes the number of latent factors, and P denotes the dimension of the samples.

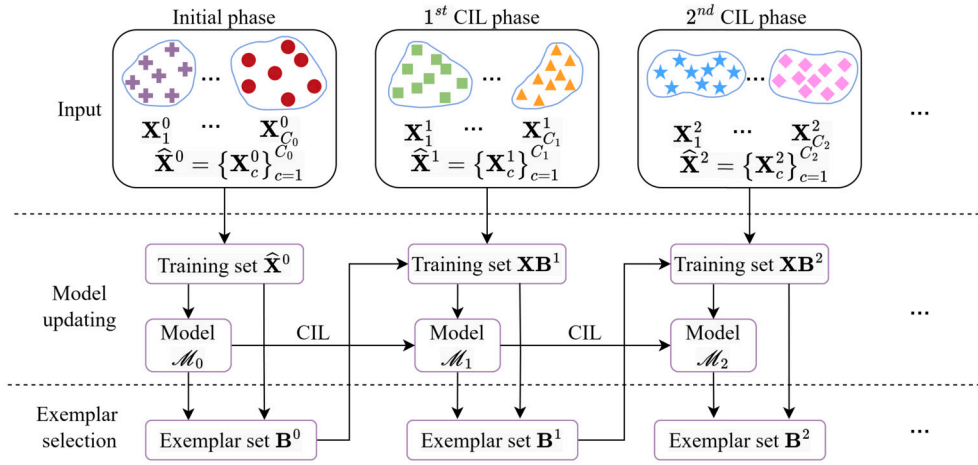


Fig. 2. The flowchart of memory-replay-based CIL method. The CIL process begins with an initial model $\mathcal{M}_0$ which is trained on C_0 initial classes. The boundary of each class in the data distribution space is represented with the blue border, and different symbols represent the data of different classes. When new class data $\hat{\mathbf{X}}^t$ arrive at time t , the model $\mathcal{M}_{t-1}$ is updated to obtain the model $\mathcal{M}_t$ with $\mathbf{XB}^t$, which is consisted of $\hat{\mathbf{X}}^t$ and the exemplar set $\mathbf{B}^{t-1}$ with $\mathbf{B}^{t-1}$ covering all old classes. After model updating, we select the exemplar set $\mathbf{B}^t$ from $\mathbf{XB}^t$ to save in the memory. The rest of CIL process is performed in the same manner.

selected at the last CIL phase is available to update the model at time t besides new class data $\hat{\mathbf{X}}^t$. For simplicity, we refer to the data containing the training set $\hat{\mathbf{X}}^t$ and the exemplar set $\mathbf{B}^{t-1}$ as $\mathbf{XB}^t$ in the following. After the model update, the exemplar set is updated by reselecting a few representative exemplars from $\mathbf{XB}^t$ for model update at the next CIL phase.

3. Our method

In this section, we develop a novel CIL-KLMES for classification based on MMFA model. In this section, we first define the model structure at time t . According to the flowchart as shown in Fig. 2, new class data $\hat{\mathbf{X}}^t$ is obtained at this moment and constitute the training set $\mathbf{XB}^t$ together with the exemplar set $\mathbf{B}^{t-1}$ to update the model. The number of all seen classes is $C^t = \sum_{m=0}^t C_m$. The model structure can be implemented via the following hierarchical Bayesian framework:

$$\mathbf{x}_i^t = \mathbf{D}^t \mathbf{s}_i^t + \boldsymbol{\mu}^{0:t} + \boldsymbol{\epsilon}_i^t \quad (7)$$

$$\mathbf{d}_k^t \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{d}_k^{t-1}}, \boldsymbol{\Sigma}_{\mathbf{d}_k^{t-1}}) \quad (8)$$

$$\mathbf{s}_{ki}^t \sim \mathcal{N}(0, (\beta_k^t)^{-1}), \beta_k^t \sim Ga(c_k^{t-1}, d_k^{t-1}) \quad (9)$$

$$\boldsymbol{\epsilon}_i^t \sim \mathcal{N}(0, (\gamma^t)^{-1}), \gamma_p^t \sim Ga(e_p^{t-1}, f_p^{t-1}) \quad (10)$$

$$y_i^t, \lambda_i^t | \mathbf{W}^t, \mathbf{s}_i^t \sim \phi(y_i^t, \lambda_i^t | \mathbf{W}^t, \mathbf{s}_i^t) \quad (11)$$

$$\begin{cases} \mathbf{w}_c^t \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_c^{t-1}}, \boldsymbol{\Sigma}_{\mathbf{w}_c^{t-1}}), & \text{if } c \text{ is an old class} \\ \mathbf{w}_c^t \sim \mathcal{N}(0, \sigma_{\mathbf{w}_c}^{-1} \mathbf{I}_{K+1}), \sigma_{\mathbf{w}_c} \sim Ga(\tau_1, \tau_2), & \text{if } c \text{ is a new class} \end{cases} \quad (12)$$

where $\mathbf{x}_i^t$ is the i th sample in $\mathbf{X}\mathbf{B}^t$, $y_i^t \in \{-1, +1\}$ is the label of it. If $y_i^t = +1$, $\mathbf{x}_i^t$ is a positive sample, else being a negative sample. $\boldsymbol{\mu}^{0:t}$ is the mean vector of all seen classes. $\mathbf{d}_k^t$ is the k th column vector of the factor loading matrix $\mathbf{D}^t \in \mathbb{R}^{P \times K}$ with $\mathbf{d}_k^t$ obeying the Gaussian distribution as $\mathbf{d}_k^t \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{d}_k^{t-1}}, \boldsymbol{\Sigma}_{\mathbf{d}_k^{t-1}})$, and $\boldsymbol{\mu}_{\mathbf{d}_k^{t-1}}, \boldsymbol{\Sigma}_{\mathbf{d}_k^{t-1}}$ are respectively the mean and covariance of the posterior distribution of $\mathbf{d}_k^{t-1}$. The superscript $t-1$ represents the variables learned at time $t-1$. $\mathbf{s}_i^t \in \mathbb{R}^K$ is the latent variable with s_{ki}^t being the k th element of $\mathbf{s}_i^t$ and $s_{ki}^t \sim \mathcal{N}(0, (\beta_k^t)^{-1})$; the precisions β_k^t obey Gamma distributions as $\beta_k^t \sim Ga(c_k^{t-1}, d_k^{t-1})$ with c_k^{t-1} and d_k^{t-1} being the hyper-parameters of the posterior distribution of β_k^{t-1} . $\epsilon_i^t \in \mathbb{R}^P$ is the noise variable with $\epsilon_i^t \sim \mathcal{N}(0, (\gamma^t)^{-1})$, and γ_p^t obeys Gamma distribution as $\gamma_p^t \sim Ga(e_p^{t-1}, f_p^{t-1})$ with γ_p^t denoting the p th diagonal element of $\boldsymbol{\gamma}^t$; analogously, e_p^{t-1} and f_p^{t-1} are the hyper-parameters of the posterior distribution of γ_p^{t-1} . In addition, $\mathbf{w}_c^t$ is the c th column vector of the weighted coefficient $\mathbf{W}^t \in \mathbb{R}^{(K+1) \times C^t}$; $\mathbf{w}_c^t$ obeys Gaussian distribution as $\mathbf{w}_c^t \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_c^{t-1}}, \boldsymbol{\Sigma}_{\mathbf{w}_c^{t-1}})$ if c corresponding to the old classes otherwise obeying Gaussian distribution as $\mathbf{w}_c^t \sim \mathcal{N}(0, \sigma_{\mathbf{w}_c}^{-1} \mathbf{I}_{K+1})$; $\boldsymbol{\mu}_{\mathbf{w}_c^{t-1}}$ and $\boldsymbol{\Sigma}_{\mathbf{w}_c^{t-1}}$ are the mean and covariance of the posterior distribution of $\mathbf{w}_c^{t-1}$ respectively. $\sigma_{\mathbf{w}_c}$ is the precision of $\mathbf{w}_c^t$ with τ_1 and τ_2 being the preset hyper-parameters. $\phi(y_i^t, \lambda_i^t | \mathbf{W}^t, \mathbf{s}_i^t)$ is the joint distribution of the label y_i^t and the augmented variable λ_i^t , which can be represented as $\phi(y_i^t, \lambda_i^t | \mathbf{W}^t, \mathbf{s}_i^t) = \frac{1}{\sqrt{2\pi\lambda_i^t}} \exp\left(-\frac{(\lambda_i^t + h_0(1-y_i^t(\mathbf{W}^t)^T \mathbf{s}_i^t))^2}{2\lambda_i^t}\right)$ with $\mathbf{s}_i^t = [\mathbf{s}_i^t; 1]$ denoting

the augmented feature vector. Thus, we have $\mathbf{x}_i^t \sim \mathcal{N}(\boldsymbol{\mu}^t, \mathbf{D}^t (\mathbf{D}^t)^T + \boldsymbol{\gamma}^t)$. According to [34], the latent variable $\mathbf{S}^t = \{\mathbf{s}_i^t\}_{i=1}^{|\mathbf{X}\mathbf{B}^t|}$ and the augmented variable $\lambda^t = \{\lambda_i^t\}_{i=1}^{|\mathbf{X}\mathbf{B}^t|}$ are local variables ξ^t , where $|\cdot|$ denotes the size of associated data set, since $\mathbf{s}_i^t$ and λ_i^t are specific to the observation $\mathbf{x}_i^t$. The factor loading matrix $\mathbf{D}^t$, the weighted coefficient $\mathbf{W}^t$, the parameters $\boldsymbol{\beta}^t = \{\beta_k^t\}_{k=1}^K$ and $\boldsymbol{\gamma}^t$ are global variables θ^t , which are shared among all data. The global variables can reflect the main characteristics of the model. For example, the factor loading matrix $\mathbf{D}^t$ is to map data into a common feature space, which reflects the projection plane. Analogous, the weighted coefficient $\mathbf{W}^t$ determines the classification boundary in this feature space. Unlike MMFA model, the posteriors of global variables trained at time $t-1$ are regarded as the priors at time t , such as the factor loading matrix $\mathbf{D}^t$, the precisions of the latent variable β_k^t , the precisions of the noise variable γ_p^t and the weighted coefficient corresponding to the old classes $\mathbf{w}_c^t$, since some key knowledge of the past classes is represented in the form of the posteriors of old model parameters.

For alleviating the CF problem, our CIL-KLMES introduces a KL constraint term to preserve the previous learned knowledge of old classes. Moreover, a few representative replay exemplars of old classes are selected from the aspects of data distribution description and classification decision boundary preservation and participate in the model update together with new class data for further alleviating the CF problem.

3.1. Learning with Kullback-Leibler constraint

For a better description of data and labels, the model parameters are inferred according to maximizing marginal likelihood [36]. The logarithm marginal likelihood is defined as:

$$\ln p(\mathbf{X}\mathbf{B}^t, \mathbf{y}^t) = \ln \int p(\mathbf{X}\mathbf{B}^t, \mathbf{y}^t, \theta^t, \xi^t) d\theta^t d\xi^t \quad (13)$$

where $\theta^t = \{\mathbf{D}^t, \boldsymbol{\beta}^t, \boldsymbol{\gamma}^t, \mathbf{W}^t\}$ and $\xi^t = \{\mathbf{S}^t, \lambda^t\}$. $\mathbf{y}^t$ is the label set of $\mathbf{X}\mathbf{B}^t$. We employ Jensen's inequality to derive a lower bound LB of the logarithm marginal likelihood. Jensen's inequality [40] states that for a convex function $f(x)$, the expected value of $f(x)$ (i.e., $E(f(x))$) is greater than or equal to $f(E(x))$, where $E(x)$ is the expected value of x . If x is a continuous variable, we have $f(\int xp(x)dx) \leq \int f(x)p(x)dx$. According to Jensen's inequality, the lower bound LB of the logarithm marginal likelihood can be expressed as:

$$LB = \int q(\theta^t, \xi^t) \ln \frac{p(\mathbf{X}\mathbf{B}^t, \mathbf{y}^t, \theta^t, \xi^t)}{q(\theta^t, \xi^t)} d\theta^t d\xi^t \quad (14)$$

In general, we maximize the lower bound to achieve maximizing the logarithm marginal likelihood. We address the CF problem by introducing a KL divergence term, aiming at preserving the previous knowledge of the past classes at the parameter level. The KL divergence [41] measures the similarity between two probability density functions. When two distributions exhibit higher similarity, the KL divergence between them is smaller. By KL constraint term, the parameters updated at two adjacent incremental phases are close, and the updated model can be prevented from deviating too much from the previous model. In detail, the KL constraint is utilized to restrict the global variables' posterior distributions learned at two adjacent incremental phases to be similar; thus, the characteristics of the previous model can be preserved. The KL constraint term can be written as:

$$\begin{aligned}
L_{KL} &= KL(q(\theta^t) \| q(\theta^{t-1})) \\
&= \int q(\theta^t) \ln \frac{q(\theta^t)}{q(\theta^{t-1})} d\theta^t
\end{aligned} \tag{15}$$

where $q(\theta^t)$ and $q(\theta^{t-1})$ are the global variables' posterior distributions learned at time t and time $t-1$ respectively; L_{KL} is constantly non-negative. Based on the above, the objective function of our CIL-KLMES method can be denoted as:

$$L(\theta^t, \xi^t) = LB - u \cdot KL(q(\theta^t) \| q(\theta^{t-1})) \tag{16}$$

where the weight value u is a tradeoff between the lower bound LB and the KL constraint term. The variational method iteratively updates each parameter alternatively until the objective function $L(\theta^t, \xi^t)$ is maximized, and the details are shown in Section 4.

However, the ability of previous knowledge preservation by KL constraint is limited. The other measure that a few representative exemplars are selected and replayed is introduced. The details of our exemplar selection method are described in the following.

3.2. Multi-strategy representative exemplar selection

For exemplar selection, the main challenge is how to select the exemplars with more representativeness. This section shows the details of our multi-strategy representative exemplar selection method. We select a few representative replay samples considering two aspects of information: data distribution description and class decision boundary preservation. The one is selecting a few samples with larger likelihood values according to maximum likelihood criterion, and the other one is selecting a few samples based on how close they are to the classification decision boundary as the samples closer to the classification decision boundary have greater influence on the classification.

Algorithm 1 Multi-strategy Representative Exemplar selection method.

Input: $\{\hat{\mathbf{X}}^0, \hat{\mathbf{X}}^1, \dots, \hat{\mathbf{X}}^t, \dots\}$, N_{dd} , N_{db}
Output: $\{\mathbf{B}^0, \mathbf{B}^1, \dots, \mathbf{B}^t, \dots\}$
1: **for** $t = 0, 1, 2, \dots$ **do**
2: Compute n_{dd}^c and n_{db}^c by $\lfloor N_{dd} / \sum_{m=0}^t C_m \rfloor$ and $\lfloor N_{db} / \sum_{m=0}^t C_m \rfloor$;
3: **if** $t = 0$ **then**
4: $\mathbf{XB}^0 = \hat{\mathbf{X}}^0$;
5: **else**
6: Construct $\mathbf{XB}^t$ with $\hat{\mathbf{X}}^t$ and $\mathbf{B}^{t-1}$;
7: **end if**
8: **for each** $\mathbf{x}_i^t \in \mathbf{XB}^t$ **do**
9: Compute lik_i^t and dis_i^t according to Eq. (17) and Eq. (18), respectively;
10: **end for**
11: Produce sort lists based on lik_i^t and dis_i^t in descending and ascending order respectively, and select first n_{dd}^c and n_{db}^c samples according to these two lists; save the selected samples and the corresponding labels as $\mathbf{B}^t$ in the memory buffer.
12: **end for**

In detail, we consider a memory buffer with a limited capacity of R samples. We assume that the exemplar set $\mathbf{B}^t$ in the memory buffer based on the two strategies are $\mathbf{B}_{dd}^t$ and $\mathbf{B}_{db}^t$ with N_{dd} and N_{db} being the sizes of them respectively, which satisfies $R = N_{dd} + N_{db}$. The subscript dd is the indicator which implies $\mathbf{B}_{dd}^t$ is the exemplar set based on the data distribution description. Analogically, the subscript db is the indicator which implies $\mathbf{B}_{db}^t$ is the exemplar set selected based on the class decision boundary preservation. Without loss of generality, $N_{dd} = N_{db}$, and each class is treated equally, i.e., we select $n_{dd}^c = \lfloor N_{dd} / \sum_{m=0}^t C_m \rfloor$ and $n_{db}^c = \lfloor N_{db} / \sum_{m=0}^t C_m \rfloor$ exemplars based on the above two strategies from each class at time t . For selecting the exemplar set $\mathbf{B}_{dd}^t$, a sorted list of the logarithm likelihoods of $\mathbf{XB}^t$ in descending order is produced for each class. The logarithm likelihood lik_i^t of $\mathbf{x}_i^t \in \mathbf{XB}^t$ can be written as

$$lik_i^t = -P/2 \ln(2\pi) + 1/2 \langle \ln \mathbf{r}^t \rangle - 1/2 \left\langle \mathbf{r}^t (\mathbf{x}_i^t - \mathbf{D}^t \mathbf{s}_i^t)^T (\mathbf{x}_i^t - \mathbf{D}^t \mathbf{s}_i^t) \right\rangle \tag{17}$$

where $\langle \cdot \rangle$ indicates the expectation. Given the sorted list, the first n_{dd}^c samples of each class and the corresponding labels are selected as $\mathbf{B}_{dd}^t$ and saved in the memory buffer. Analogously, for selecting the exemplar set $\mathbf{B}_{db}^t$, a sorted list of the distance to the classification decision boundary of $\mathbf{XB}^t$ in ascending order is produced for each class. The distance dis_i^t from $\mathbf{x}_i^t \in \mathbf{XB}^t$ to its corresponding classification decision boundary can be expressed as

$$dis_i^t = \left| \left\langle (\mathbf{w}_c^t)^T \right\rangle \left\langle \tilde{\mathbf{s}}_i^t \right\rangle \right| \tag{18}$$

where $\mathbf{w}_c^t$ is the weighted coefficient corresponding to the class of $\mathbf{x}_i^t$. Given the sorted list, the first n_{db}^c samples of each class and the corresponding labels are selected as $\mathbf{B}_{db}^t$ and saved in the memory buffer. Based on the above, the exemplar set $\mathbf{B}^t$ is composed of $\mathbf{B}_{dd}^t$ and $\mathbf{B}_{db}^t$ at time t . After that, $\mathbf{XB}^{t+1}$ is composed of $\hat{\mathbf{X}}^{t+1}$ and the exemplar set $\mathbf{B}^t$ for updating the model when $\hat{\mathbf{X}}^{t+1}$ arrives at next time, and then the above exemplar selection method is repeated to update the exemplar set. In detail, the number of all seen classes changes to $C^{t+1} = \sum_{m=0}^{t+1} C_m$ with n_{dd}^c and n_{db}^c changing to $n_{dd}^{c+1} = \lfloor N_{dd} / \sum_{m=0}^{t+1} C_m \rfloor$ and $n_{db}^{c+1} = \lfloor N_{db} / \sum_{m=0}^{t+1} C_m \rfloor$ when new classes $\hat{\mathbf{X}}^{t+1}$

are added. For each old class, we choose n_{dd}^c exemplars from the old exemplar set $\mathbf{B}_{dd}^t$ and n_{db}^c exemplars from the old exemplar set $\mathbf{B}_{db}^t$ respectively according to our proposed exemplar selection strategy. For new classes, we choose $n_{dd}^c + n_{db}^c$ exemplars per class from $\hat{\mathbf{X}}^{t+1}$ according to our proposed exemplar selection method. Thus, the updated exemplar set $\mathbf{B}^{t+1}$ can be obtained. Combining multiple exemplar selection strategies can further ensure the integrality of the key knowledge preservation of the past classes, thus effectively alleviating the CF problem by replaying the exemplar set.

By this selection method, the exemplar set is constantly updated by adding some representative exemplars from new classes and deleting some representative exemplars from the current exemplar set with the increase of classes. Thus, it can ensure the size of the total memory buffer is not more than R . The multi-strategy representative exemplar selection method is summarized as Algorithm 1.

4. Parameter optimization and classification

This section presents the parameter optimization and classification at each CIL phase. In the training phases, our goal is to calculate the posterior distributions of the model parameters based on $\mathbf{XB}^t$ at time t . The parameters can be inferred via SVI algorithm, of which the main idea is to combine SGD algorithm and variational inference. SVI allows data to be processed in mini-batches, and the parameters can be updated iteratively. In SVI, the optimization equations of the local variables ξ^t are similar to those solved by traditional variational Bayesian algorithm [36–38]. The global variables θ^t are updated using stochastic natural gradient ascent. At iteration l (corresponding to the l th mini-batch), the update form is:

$$\theta^{t,l} = \theta^{t,l-1} + \rho_l \tilde{\nabla}_{\theta} L(\theta^t, \xi^t) \quad (19)$$

where $\theta^{t,l} = \{\mathbf{D}^{t,l}, \beta^{t,l}, \gamma^{t,l}, \mathbf{W}^{t,l}\}$ denotes the estimation at iteration l ; ρ_l is the learning step; $\tilde{\nabla}_{\theta} L(\theta^t, \xi^t)$ is the natural gradient of the objective function with respect to θ^t . According to the derivation in [34], the updates can take the form of:

$$\theta^{t,l} = (1 - \rho_l) \theta^{t,l-1} + \rho_l \hat{\theta}^{t,l} \quad (20)$$

where the intermediate parameters $\hat{\theta}^{t,l} = \{\hat{\mathbf{D}}^{t,l}, \hat{\beta}^{t,l}, \hat{\gamma}^{t,l}, \hat{\mathbf{W}}^{t,l}\}$ are computed using the mini-batch corresponding to iteration l . In detail, the optimization equations at iteration l can be derived as follows.

1) $s_i^{t,l}$ is a local variable, and the optimization equations of $s_{ki}^{t,l}$ are as follows:

$$q(s_{ki}^{t,l}) = \mathcal{N}(\mu_{s_{ki}^{t,l}}, \Sigma_{s_{ki}^{t,l}}) \quad (21)$$

$$\mu_{s_{ki}^{t,l}} = \Sigma_{s_{ki}^{t,l}} \left(\langle \hat{\mathbf{d}}_k^{t,l} \rangle^T \langle \hat{\gamma}^{t,l} \rangle \Lambda_i^{t,l} + \sum_{c=1}^{C^t} (h_0 y_{ci}^{t,l} \langle \hat{w}_{kc}^{t,l} \rangle + h_0^2 O_{ci}^{t,l} \langle \hat{w}_{kc}^{t,l} \rangle) \right) \quad (22)$$

$$\Sigma_{s_{ki}^{t,l}} = \left(\langle (\hat{\mathbf{d}}_k^{t,l})^T \hat{\gamma}^{t,l} \hat{\mathbf{d}}_k^{t,l} \rangle + \langle \hat{\beta}_k^{t,l} \rangle + \sum_{c=1}^{C^t} h_0^2 \langle (\lambda_{ci}^{t,l})^{-1} \rangle \langle (\hat{w}_{kc}^{t,l})^2 \rangle \right)^{-1} \quad (23)$$

where $\hat{w}_{kc}^{t,l}$ is the k th element of $\hat{\mathbf{w}}_c^{t,l}$ with $\hat{\mathbf{w}}_c^{t,l}$ being the c th column vector of $\hat{\mathbf{W}}^{t,l}$; $\lambda_{ci}^{t,l}$ is the augmented variable corresponding to class c at iteration l . Moreover, $y_{ci}^{t,l}$ is the label of $\mathbf{x}_i^t$ corresponding to class c , i.e., if $\mathbf{x}_i^t$ belongs to class c , $y_{ci}^{t,l} = +1$, otherwise

$$y_{ci}^{t,l} = -1. \Lambda_i^{t,l} = \mathbf{x}_i^t - \sum_{m=1, m \neq k}^K \langle s_{mi}^{t,l} \rangle \langle \hat{\mathbf{d}}_m^{t,l} \rangle \text{ and } O_{ci}^{t,l} = \left\langle (\lambda_{ci}^{t,l})^{-1} \right\rangle \left(y_{ci}^{t,l} - \sum_{m=1, m \neq k}^{K+1} \langle \hat{s}_{mi}^{t,l} \rangle \langle \hat{w}_{mc}^{t,l} \rangle \right).$$

2) For the augmented variable $\lambda_i^{t,l}$, we have

$$\begin{aligned} p(\lambda_i^{t,l} | \hat{\mathbf{W}}^{t,l}, s_i^{t,l}) &\propto \frac{1}{\sqrt{2\pi\lambda_i^{t,l}}} \exp \left(-\frac{\left(\lambda_i^{t,l} + h_0 \left(1 - y_i^t (\hat{\mathbf{W}}^{t,l})^T \hat{\mathbf{s}}_i^{t,l} \right) \right)^2}{2\lambda_i^{t,l}} \right) \\ &= \mathcal{GIG} \left(\lambda_i^{t,l}; \frac{1}{2}, 1, h_0^2 \left(1 - y_i^t (\hat{\mathbf{W}}^{t,l})^T \hat{\mathbf{s}}_i^{t,l} \right)^2 \right) \end{aligned} \quad (24)$$

where $\mathcal{GIG}(x; g, a, b) = \bar{C}(g, a, b) x^{g-1} \exp \left(-\frac{1}{2} (bx^{-1} + ax) \right)$ is a generalized inverse Gaussian distribution [42] with $\bar{C}(g, a, b)$ denoting a normalization constant. Then, $(\lambda_{ci}^{t,l})^{-1}$ obeys an inverse Gaussian distribution:

$$q \left((\lambda_{ci}^{t,l})^{-1} \right) \sim \mathcal{IG} \left((\lambda_{ci}^{t,l})^{-1}; 1 / \left(h_0 \left| 1 - y_{ci}^t \langle (\hat{\mathbf{w}}_c^{t,l})^T \rangle \langle \hat{s}_i^{t,l} \rangle \right| \right), 1 \right) \quad (25)$$

where $\mathcal{IG}(x; a, b) = \sqrt{\frac{b}{2\pi x^3}} \exp \left(-\frac{b(x-a)^2}{2a^2 x} \right)$ for $a > 0, b > 0$.

3) Loading matrix $\mathbf{D}^t$ is a global variable, and the optimization equations of $\mathbf{d}_k^t$ at iteration l , $\mathbf{d}_k^{t,l} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{d}_k^{t,l}}, \boldsymbol{\Sigma}_{\mathbf{d}_k^{t,l}})$, are as follows:

$$\hat{\boldsymbol{\mu}}_{\mathbf{d}_k^{t,l}} = \hat{\boldsymbol{\Sigma}}_{\mathbf{d}_k^{t,l}} \left(\frac{|\mathbf{XB}^t|}{(1+u)M} \langle \hat{\mathbf{y}}^{t,l} \rangle \sum_{i=1}^M \Lambda_i^{t,l} \langle s_{ki}^{t,l} \rangle + \boldsymbol{\Sigma}_{\mathbf{d}_k^{t-1}}^{-1} \boldsymbol{\mu}_{\mathbf{d}_k^{t-1}} \right) \quad (26)$$

$$\hat{\boldsymbol{\Sigma}}_{\mathbf{d}_k^{t,l}} = \left(\frac{|\mathbf{XB}^t|}{(1+u)M} \langle \hat{\mathbf{y}}^{t,l} \rangle \sum_{i=1}^M \left\langle \left(s_{ki}^{t,l} \right)^2 \right\rangle + \boldsymbol{\Sigma}_{\mathbf{d}_k^{t-1}}^{-1} \right)^{-1} \quad (27)$$

$$\boldsymbol{\mu}_{\mathbf{d}_k^{t,l}} = (1 - \rho_l) \boldsymbol{\mu}_{\mathbf{d}_k^{t,l-1}} + \rho_l \hat{\boldsymbol{\mu}}_{\mathbf{d}_k^{t,l}} \quad (28)$$

$$\boldsymbol{\Sigma}_{\mathbf{d}_k^{t,l}} = (1 - \rho_l) \boldsymbol{\Sigma}_{\mathbf{d}_k^{t,l-1}} + \rho_l \hat{\boldsymbol{\Sigma}}_{\mathbf{d}_k^{t,l}} \quad (29)$$

where M is the size of each mini-batch, and $\Lambda_i^{t,l}$ is the same as in Eq. (22).

4) The optimization equations of the global parameter β_k^t at iteration l are as follows:

$$\hat{c}_k^{t,l} = c_k^{t-1} + \frac{|\mathbf{XB}^t|}{2(1+u)} \quad (30)$$

$$\hat{d}_k^{t,l} = d_k^{t-1} + \frac{|\mathbf{XB}^t|}{2(1+u)M} \sum_{i=1}^M \left\langle \left(s_{ki}^{t,l} \right)^2 \right\rangle \quad (31)$$

$$c_k^{t,l} = (1 - \rho_l) c_k^{t,l-1} + \rho_l \hat{c}_k^{t,l} \quad (32)$$

$$d_k^{t,l} = (1 - \rho_l) d_k^{t,l-1} + \rho_l \hat{d}_k^{t,l} \quad (33)$$

5) The optimization equations of global parameter γ_p^t at iteration l are as follows:

$$\hat{e}_p^{t,l} = e_p^{t-1} + \frac{|\mathbf{XB}^t|}{2(1+u)} \quad (34)$$

$$\hat{f}_p^{t,l} = f_p^{t-1} + \frac{|\mathbf{XB}^t|}{2(1+u)M} \sum_{i=1}^M \left\langle \left(x_{pi}^t - \tilde{\mathbf{d}}_p^{t,l} s_i^{t,l} \right)^2 \right\rangle \quad (35)$$

$$e_p^{t,l} = (1 - \rho_l) e_p^{t,l-1} + \rho_l \hat{e}_p^{t,l} \quad (36)$$

$$f_p^{t,l} = (1 - \rho_l) f_p^{t,l-1} + \rho_l \hat{f}_p^{t,l} \quad (37)$$

where x_{pi}^t is the p th element of $\mathbf{x}_i^t$; $\tilde{\mathbf{d}}_p^{t,l}$ is the p th row of the loading matrix $\hat{\mathbf{D}}^{t,l}$.

6) $\mathbf{w}_c^t$ is the weighted coefficient corresponding to the old class c , and we have:

$$\hat{\boldsymbol{\mu}}_{\mathbf{w}_c^{t,l}} = \hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} \left(\frac{|\mathbf{XB}^t|}{(1+u)M} \sum_{i=1}^M \boldsymbol{\Theta}_{ci}^{t,l} + \boldsymbol{\Sigma}_{\mathbf{w}_c^{t-1}}^{-1} \boldsymbol{\mu}_{\mathbf{w}_c^{t-1}} \right) \quad (38)$$

$$\hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} = \left(\frac{|\mathbf{XB}^t|}{(1+u)M} \sum_{i=1}^M h_0^2 \left\langle \left(\lambda_{ci}^{t,l} \right)^{-1} \right\rangle \left\langle \tilde{\mathbf{s}}_i^{t,l} \left(\tilde{\mathbf{s}}_i^{t,l} \right)^T \right\rangle + \boldsymbol{\Sigma}_{\mathbf{w}_c^{t-1}}^{-1} \right)^{-1} \quad (39)$$

$$\boldsymbol{\mu}_{\mathbf{w}_c^{t,l}} = (1 - \rho_l) \boldsymbol{\mu}_{\mathbf{w}_c^{t,l-1}} + \rho_l \hat{\boldsymbol{\mu}}_{\mathbf{w}_c^{t,l}} \quad (40)$$

$$\boldsymbol{\Sigma}_{\mathbf{w}_c^{t,l}} = (1 - \rho_l) \boldsymbol{\Sigma}_{\mathbf{w}_c^{t,l-1}} + \rho_l \hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} \quad (41)$$

where $\boldsymbol{\Theta}_{ci}^{t,l} = h_0 y_{ci}^t \langle \tilde{\mathbf{s}}_i^{t,l} \rangle + h_0^2 y_{ci}^{t,l} \left\langle \left(\lambda_{ci}^{t,l} \right)^{-1} \right\rangle \langle \tilde{\mathbf{s}}_i^{t,l} \rangle$. In addition, it should be noted that the optimization equations of some parameters are different from Eq. (38) – Eq. (41) when new classes firstly occur. The intermediate parameters of some variables related to the classifier corresponding to new classes, including $\mathbf{w}_c^t, \sigma_{\mathbf{w}_c^t}$, are optimized as follows.

7) The update equations of $\mathbf{w}_c^t$ corresponding to new classes are as follows:

$$\hat{\boldsymbol{\mu}}_{\mathbf{w}_c^{t,l}} = \hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} \left(\frac{|\mathbf{XB}^t|}{M} \sum_{i=1}^M \left(h_0 y_{ci}^{t,l} \langle \tilde{\mathbf{s}}_i^{t,l} \rangle + h_0^2 y_{ci}^{t,l} \left\langle \left(\lambda_{ci}^{t,l} \right)^{-1} \right\rangle \langle \tilde{\mathbf{s}}_i^{t,l} \rangle \right) \right) \quad (42)$$

$$\hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} = \left(\frac{|\mathbf{XB}^t|}{M} \sum_{i=1}^M h_0^2 \left\langle \left(\lambda_{ci}^{t,l} \right)^{-1} \right\rangle \left\langle \tilde{\mathbf{s}}_i^{t,l} \left(\tilde{\mathbf{s}}_i^{t,l} \right)^T \right\rangle + \langle \sigma_{\mathbf{w}_c} \rangle \mathbf{I}_{K+1} \right)^{-1} \quad (43)$$

$$\boldsymbol{\mu}_{\mathbf{w}_c^{t,l}} = (1 - \rho_l) \boldsymbol{\mu}_{\mathbf{w}_c^{t,l-1}} + \rho_l \hat{\boldsymbol{\mu}}_{\mathbf{w}_c^{t,l}} \quad (44)$$

$$\boldsymbol{\Sigma}_{\mathbf{w}_c^{t,l}} = (1 - \rho_l) \boldsymbol{\Sigma}_{\mathbf{w}_c^{t,l-1}} + \rho_l \hat{\boldsymbol{\Sigma}}_{\mathbf{w}_c^{t,l}} \quad (45)$$

8) The optimization equations of σ_{w_c} corresponding to new classes are as follows:

$$\hat{\tau}_1^{t,l} = \tau_1 + \frac{K+1}{2} \quad (46)$$

$$\hat{\tau}_2^{t,l} = \tau_2 + \frac{1}{2} \left\langle (\hat{w}_c^{t,l})^T \hat{w}_c^{t,l} \right\rangle \quad (47)$$

$$\tau_1^{t,l} = (1 - \rho_l) \tau_1^{t,l-1} + \rho_l \hat{\tau}_1^{t,l} \quad (48)$$

$$\tau_2^{t,l} = (1 - \rho_l) \tau_2^{t,l-1} + \rho_l \hat{\tau}_2^{t,l} \quad (49)$$

In the experiments of this paper, the algorithm is terminated when the variation rate of the objective function $L(\theta^t, \xi^t)$ between two iterations is smaller than 10^{-6} . The training process of CIL-KLMES is summarized in Algorithm 2.

In the classification stage, the class label of a test sample $\mathbf{x}^*$ can be predicted with the classification criteria of support vector machine (SVM), and the prediction is given by the following steps:

1) The latent variable $\mathbf{s}^*$ of $\mathbf{x}^*$ at time t is shown as

$$q(\mathbf{s}_k^*) = \mathcal{N}(\mu_{s_k^*}, \Sigma_{s_k^*}) \quad (50)$$

$$\mu_{s_k^*} = \Sigma_{s_k^*} \left(\left\langle (\mathbf{d}_k^t)^T \right\rangle \left\langle \gamma^t \right\rangle \left(\mathbf{x}^* - \mu^{0:t} - \sum_{\substack{m=1 \\ (m \neq k)}}^K \left\langle s_m^* \right\rangle \left\langle \mathbf{d}_m^t \right\rangle \right) \right) \quad (51)$$

$$\Sigma_{s_k^*} = \left(\left\langle (\mathbf{d}_k^t)^T \right\rangle \left\langle \gamma^t \right\rangle \left\langle \mathbf{d}_k^t \right\rangle + \left\langle \beta_k^t \right\rangle \right)^{-1} \quad (52)$$

2) Computing the label $\hat{y}^*$ of $\mathbf{x}^*$ at time t based on the following equations:

$$\hat{y}^* = \text{sign} \left((\mathbf{W}^t)^T \tilde{\mathbf{s}}^* \right) \quad (53)$$

where $\tilde{\mathbf{s}}_i^* = [\mathbf{s}_i^*; 1]$ is the augmented variable of $\mathbf{s}^*$. If $\hat{y}^* = +1$, $\mathbf{x}^*$ is classified as a positive sample whereas as a negative sample ($\hat{y}^* = -1$).

Algorithm 2 Training process of CIL-KLMES.

Input: $\{\hat{\mathbf{X}}^0, \hat{\mathbf{X}}^1, \dots, \hat{\mathbf{X}}^t, \dots\}$, N_{dd} , N_{db} , $a_0, b_0, c_0, d_0, e_0, f_0, \tau_1, \tau_2, h_0$

Output: $\mathcal{M}_t = \{\mathbf{D}^t, \mathbf{S}^t, \beta^t, \gamma^t, \mathbf{W}^t, \lambda^t\}$ and $\mathbf{B}^t$

```

1: for  $t = 1, 2, \dots$  do
2:   Construct  $\mathbf{XB}^t$ , initialize  $v = 0$  and  $L(v) = -\infty$ ;
3:   for each mini-batch of  $\mathbf{XB}^t$  do
4:     Estimate the local and global variables via Eq. (19) – Eq. (49);
5:      $v = v + 1$ ;
6:     Compute the objective function  $L(v)$ ;
7:     if  $(L(v) - L(v-1)) / L(v-1) < 10^{-6}$  then
8:       Break;
9:     end if
10:   end for
11:   Update the exemplar set  $\mathbf{B}^t$  by Algorithm 1;
12: end for

```

5. Experimental results

5.1. Data introduction and experimental settings

5.1.1. UCI datasets

We firstly evaluate the proposed method on UCI datasets, which can be found at the UCI Machine Learning Repository.

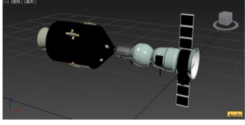
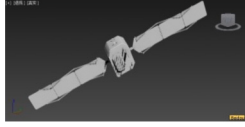
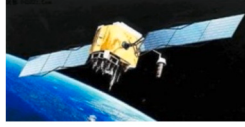
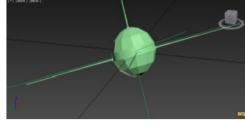
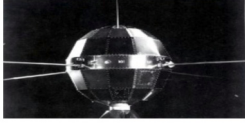
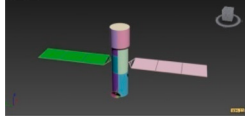
The Human Activity Recognition Using Smartphones Data Set (HARUS dataset) proposed by Anguita et al. [43] is derived from 30 volunteers' activities of daily life recorded by a waist-mounted smartphone that has embedded inertial sensors. The dataset contains three postures and three activities (standing, sitting, lying, walking, walking downstairs and walking upstairs) with the data dimension being 561. The goal is to classify 6 activities. The dataset has been partitioned into two sets, where 7352 samples are used as training data and the rest 2947 samples are as testing data.

The Gas Sensor Array Drift (GSA) dataset [44] includes six different gases (i.e., ethanol, ethylene, ammonia, acetaldehyde, acetone and toluene) collected from 16 chemical gas sensors, where 12516 samples are used as training data and the remaining 1394 samples are as testing data. Table 1 summarizes the data information about these 2 UCI datasets.

Table 1
Description of 2 UCI datasets.

Dataset	Number of classes	Data dimension	Number of training data	Number of testing data
HARUS	6	561	7352	2947
GSA	6	128	12516	1394

Table 2
Parameters of radar and five satellite targets in the experiments.

Radar Parameters	Center Frequency		16.67 GHz		
	Pulse Repetition Frequency		10 Hz		
	Sampling Frequency after Dechirp		10 MHz		
	Bandwidth		2 GHz		
Radar Location	39.9°N 116.4°E				
CAD Models	Real Images	Satellite Targets	Length(m)	Width(m)	Height(m)
		Apollo	21.6	9.6	5.1
		BeiDou	9.4	1.2	2
		DongFangHong	5.5	5.5	2
		KH-11	30.7	4	12
		Landsat7	7.6	2.7	3.9

5.1.2. Simulated satellite HRRP dataset

In space target surveillance systems, satellite target recognition is of significant importance for the space activity, orbit security, attack interception and some other space surveillance tasks. In satellite target recognition, new satellite targets enter the orbit continuously; therefore, CIL is more applicable for satellite target recognition. Given that the real measured satellite HRRP data cannot be shared publicly, we utilize electromagnetic simulation software FEKO [32] to simulate HRRP data of five satellite targets. The parameters of radar and five satellite targets are listed in Table 2, and the azimuth and pitch angles of the radar line-of-sight (LOS) adopt those of the actual satellite targets. Based on these parameters, we simulate the HRRP data of each target with various orbits. An HRRP [1–5] is composed of the amplitude of coherent summations of complex returns from target scatterers in each range cell, and it provides rich target information along the radar line-of-sight, such as scatterer distribution and structural characteristics.

Due to the change of orbital parameters caused by orbital transfer, the azimuth angle and pitch angles exhibit certain differences before and after orbital transfer, leading to certain differences in the data distribution. Consequently, to test the generalization performance of the classification method, we utilize data collected before orbital transfer as the training data and data collected after orbital transfer as the testing data in our experiments. In detail, a total of 20080 HRRP samples from Apollo, 14087 from BeiDou, 14988 from DongFangHong, 16197 from KH-11, and 17714 from Landsat7 are used as training data with the dimension being 512. The testing data of each target contains 1000 HRRP samples.

5.1.3. Split CIFAR-10 dataset

Split CIFAR-10 is constructed by splitting the CIFAR-10 dataset [45] into 5 different phases with nonoverlapping classes and 2 classes in each phase. Each class has 5000 32×32 RGB images for training and 1000 images for testing. For evaluating our method

on Split CIFAR-10, the histogram of oriented gradient (HOG) [46] feature vectors and HSV (Hue, Saturation, Value) color features [47] are extracted and concatenated to change two-dimensional image data into one-dimensional vectors.

In our experiments, we select two classes as initial classes and add one new class every time in CIL scenario for the first three datasets. For Split CIFAR-10, each phase has 2 classes. The experiments are performed 10 times, and the order of classes is random in each experiment. The results, including mean values and standard deviations, are obtained through t-test over the 10 experiments with p-value being less than 0.05, where the standard deviations are presented by error bars to describe the uncertainty caused by random class order. The hyperparameters of CIL-KLMES are in Gamma distribution, we usually assign small values to them to reduce the influence of priors on the estimations of model parameters. That is, the parameter estimations (the posteriors) are entirely up to the observed data (the likelihoods). This avoids the inaccurate parameter estimations caused by unreasonable hyperparameter settings. Therefore, $a_0, b_0, c_0, d_0, e_0, f_0, \tau_1$ and τ_2 are set to 10^{-6} , like [5]. For different datasets, the setting of the positive tuning parameter h_0 is different, which is chosen from $\{0.1, 1, 10\}$, like [6]. For the above datasets, h_0 are 1, 1, 0.1 and 1, respectively. The setting of learning step is $\rho_l = (l + l_0)^{-\omega}$ ($l_0 = 1, \omega = 0.5$), which refers to [34]. We use cross-validation [48] to dynamically search the optimal u , which controls the effect of the KL constraint term. In detail, a 5-fold cross validation is employed to split the training set $\mathbf{X}B'$ into training and validating splits, and then we determine the optimal u in the range of $\{0.2, 0.4, 0.6, 0.8, 1\}$. Besides, the selection ratio of the exemplar set to all training samples κ is about 0.1 for the first three datasets, i.e., the sizes of the exemplar set are 740, 1200 and 8000 respectively. For Split CIFAR-10, the size of the exemplar set is 500. In the testing process, the updated model is evaluated on the testing data of all seen classes. All the following experiments are performed with non-optimized MATLAB codes, on a Dell PC with 16 GB RAM and intel i7 CPU@3.40 GHz.

5.2. Performance comparison

In this subsection, to verify the effectiveness of our CIL-KLMES, we first evaluate the classification performance of CIL-KLMES on the above datasets by comparing with CIL-naive (baseline), which directly utilizes new class data to update the old model without taking any measures to alleviate the CF problem. Moreover, some existing CIL methods, including LwM, CIELM, FDARF, DCPOC and CBesIL, are utilized for comparison. Besides these CIL methods, we also compare our CIL-KLMES with the corresponding traditional offline MMFA model (upper bound). The classification results of these methods are shown in Fig. 3.

Fig. 3 depicts the variations of classification accuracies of the above-mentioned methods with the arrival of new classes on the datasets. The horizontal axis is the number of all seen classes at each CIL phase, and the vertical axis represents the variation of the classification accuracy. Note that the 1st value of the classification accuracy corresponds to testing the initial model $\mathcal{M}_0$. Since more and more knowledge learned from old classes is forgotten with the increase of new classes, i.e., the aggravation of CF problem, the classification accuracies of CIL methods decrease. The degree of classification performance degradation depends on the decline in classification accuracy of old classes and the classification accuracy of new classes. For old classes, the degree of forgetting is uncertain in different incremental phase. For example, the degree of forgetting is affected by the correlation between old and new classes. If the correlation between the old and the new classes is high, the knowledge learned from the new classes can adapt to the old classes, and the forgetting degree of the old classes may be relatively small; On the contrary, the degree of forgetting of the old class will be large. For new classes, the classification accuracy on new classes is unknown and depends on the degree of difficulty in the classification of the new class. These two factors result in the speed of decline in classification accuracy being uncertain. We also find that our CIL-KLMES shows better performance than other CIL methods from beginning to end and has less performance difference from MMFA model, which demonstrate the effectiveness of our method to alleviate CF problem. Since LwM and CIELM alleviate CF problem only by constraining the parameter updates, the preservation of the previous learned knowledge is insufficient, so that the model easily biases to describe new classes. This is the main reason that the classification accuracies of LwM and CIELM are lower than our CIL-KLMES. FDARF and DCPOC do not directly store the samples but generate pseudo instances, leading to more information loss. The exemplar selection method of CBesIL is only based on local geometrical information, the knowledge preservation is incomplete. Hence, the exemplars are less representative for the model update in later CIL phases. Therefore, our CIL-KLMES outperforms FDARF, DCPOC and CBesIL. In addition, since our CIL-KLMES only stores and replays a few representative exemplars, a little knowledge learned from old classes is still forgotten compared with traditional offline MMFA model; thus, the classification accuracy of our CIL-KLMES is lower than MMFA model.

We also employ Average Forgetting (AF), Forward Transfer (FWT) and Backward Transfer (BWT) [24,49] as the criteria to further evaluate the performance of our method with other CIL methods. Specially, FWT and BWT values are usually negative. FWT measures the influence to new classes of preserving the knowledge of old classes, while BWT evaluates the influence to old classes after learning new classes. For FWT, to maintain the model performance on the old classes, the flexibility of adaptation to the new classes may be sacrificed, leading to negative FWT values. Larger FWT indicates the model has more flexibility of adaptation to the new classes. For BWT, the performance on old classes generally deteriorates after learning new classes due to CF problem, resulting in negative BWT values. Large negative BWT indicates a severe degree of CF. Table 3 shows the results of AF, FWT and BWT on 4 different datasets. As we can observe, our method achieves the best performances in almost all cases, which indicates that our method suffers less forgetting and has higher flexibility.

Moreover, as a CIL method, the memory requirement and the time consumption are significant to evaluate the practicability in real applications. In Fig. 4, we analyze the memory costs of CIL-KLMES and MMFA model with the arrival of new classes on the above datasets. Here, the memory costs refer to the space occupied by the exemplar set and the model parameters of our CIL-KLMES, while MMFA model needs to store all data of all seen classes besides the related model parameters. Thus, the memory costs of CIL-KLMES are less than that of MMFA model. In addition, since the size of the exemplar set is fixed, the memory costs of our CIL-KLMES are

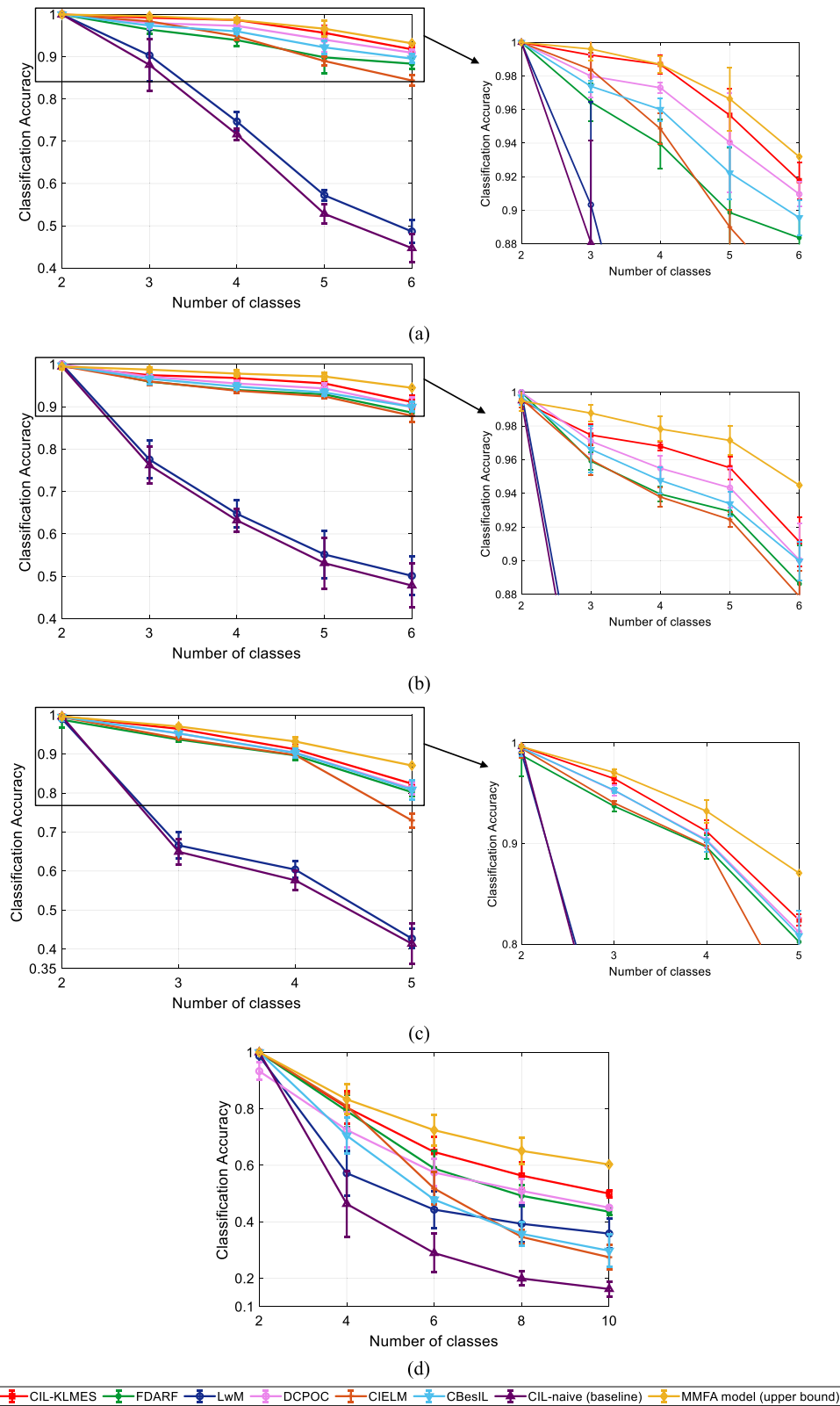


Fig. 3. The variation of classification accuracies of different methods with the arrival of new classes on the 4 datasets: (a) HARUS dataset, (b) GSA dataset, (c) simulated satellite HRRP dataset, and (d) Split CIFAR-10.

Table 3

AF, FWT and BWT on 4 datasets. The best results are marked in boldface and the second best results are marked in underlined.

Method	HARUS			GSA	FWT ↑	BWT ↑
	AF ↓	FWT ↑	BWT ↑	AF ↓	FWT ↑	BWT ↑
LwM	36.02±2.10	-12.48±1.23	-36.50±1.92	41.17±1.24	-18.37±0.92	-42.02±1.08
CIELM	7.38±0.69	-10.50±1.46	-8.39±0.34	7.25±0.82	-8.88±1.62	-8.61±1.63
FDARF	<u>4.40</u> ±1.45	-9.47±1.11	-9.15±1.31	4.15±0.69	-10.00±1.18	<u>-6.22</u> ±0.34
DCPOC	7.34±0.49	<u>-5.22</u> ±1.38	-8.35±0.70	5.75±0.89	-4.24 ±0.70	-8.77±0.75
CBesIL	5.07±1.30	-9.83±1.12	<u>-7.84</u> ±0.89	<u>4.11</u> ±0.74	-9.55±1.13	-7.03±0.96
CIL-KLMES	3.31 ±0.21	-3.71 ±0.67	-3.34 ±0.44	2.32 ±0.35	<u>-5.89</u> ±0.80	-3.78 ±0.43

Method	satellite HRRP			Split CIFAR-10		
	AF ↓	FWT ↑	BWT ↑	AF ↓	FWT ↑	BWT ↑
LwM	41.37±1.17	-17.33±0.95	-41.85±1.26	46.72±0.89	-18.19±1.04	-47.08±0.92
CIELM	10.47±0.96	-14.53±1.08	-10.82±0.98	57.92±1.88	<u>-9.41</u> ±0.71	-64.92±2.27
FDARF	8.02±0.76	-11.25±0.82	<u>-7.83</u> ±0.54	50.56±2.23	-10.28±0.89	-58.28±2.23
DCPOC	9.52±0.81	-9.94 ±1.12	-9.07±0.95	18.36±0.65	-12.75±0.74	-19.43±0.53
CBesIL	<u>7.52</u> ±0.69	-18.14±0.77	-8.23±0.52	51.85±0.81	-16.25±1.11	-52.35±0.76
CIL-KLMES	4.56 ±0.41	<u>-10.52</u> ±0.86	-5.82 ±0.50	7.19 ±0.48	-6.81 ±0.85	-8.47 ±0.53

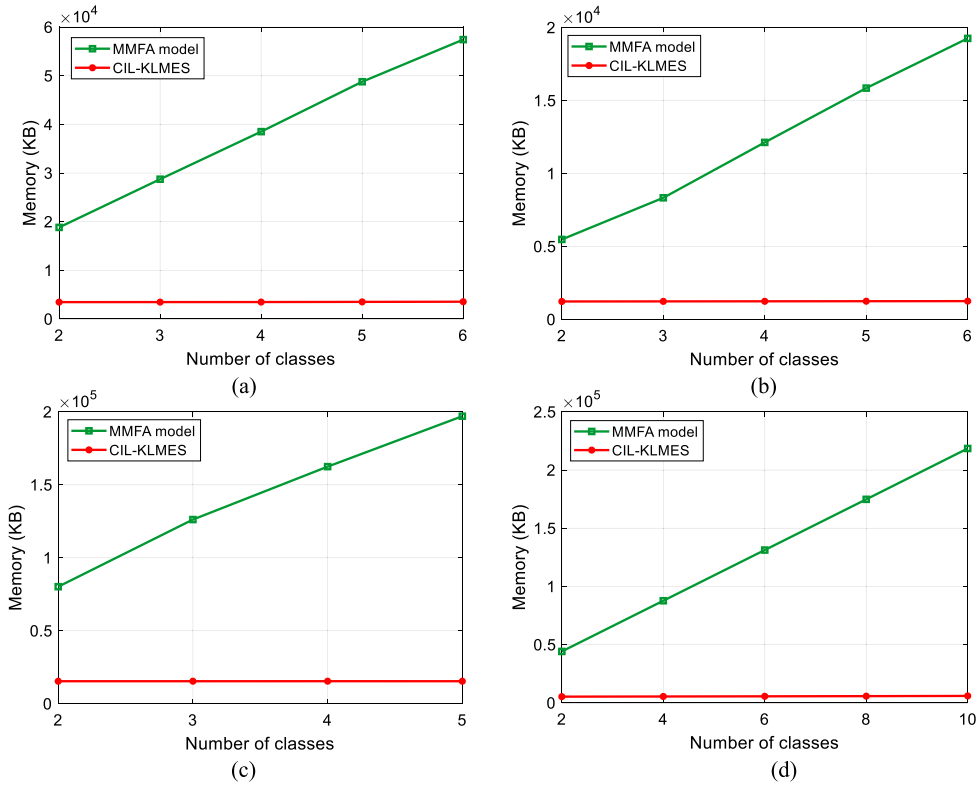


Fig. 4. The variation of memory requirement with the arrival of new classes on the 4 datasets: (a) HARUS dataset, (b) GSA dataset, (c) simulated satellite HRRP dataset, and (d) Split CIFAR-10.

basically unchanged with the arrival of new classes on each dataset. New model parameters introduced by new classes, such as the weighted coefficient corresponding to new classes, lead to small fluctuations of the memory costs.

In the following, we provide a time complexity analysis and the comparison of the time consumption of CIL-KLMES and MMFA model. According to definition, the time complexity of an algorithm is dependent on the highest order term of the times of operations. At any time t ($t \geq 1$), the time complexity of CIL-KLMES is $O(E'N_tPK^2)$ where E' is the number of iterations required for model convergence. N_t is the number of $\mathbf{X}B^t$. For MMFA model, the time complexity is $O(ENPK^2)$, where E is the number of iterations required for model convergence; N is the number of all training data until time t ($N > N_t$). Since our CIL-KLMES updates the model based on the previous trained model with $\mathbf{X}B^t$ rather than training from scratch with all data obtained until time t , we have $E > E'$, leading to $ENPK^2 > E'N_tPK^2$. That is to say, our method has a smaller time complexity compared with MMFA model. The comparison of the time consumption of CIL-KLMES and MMFA model with the arrival of new classes on the above datasets is

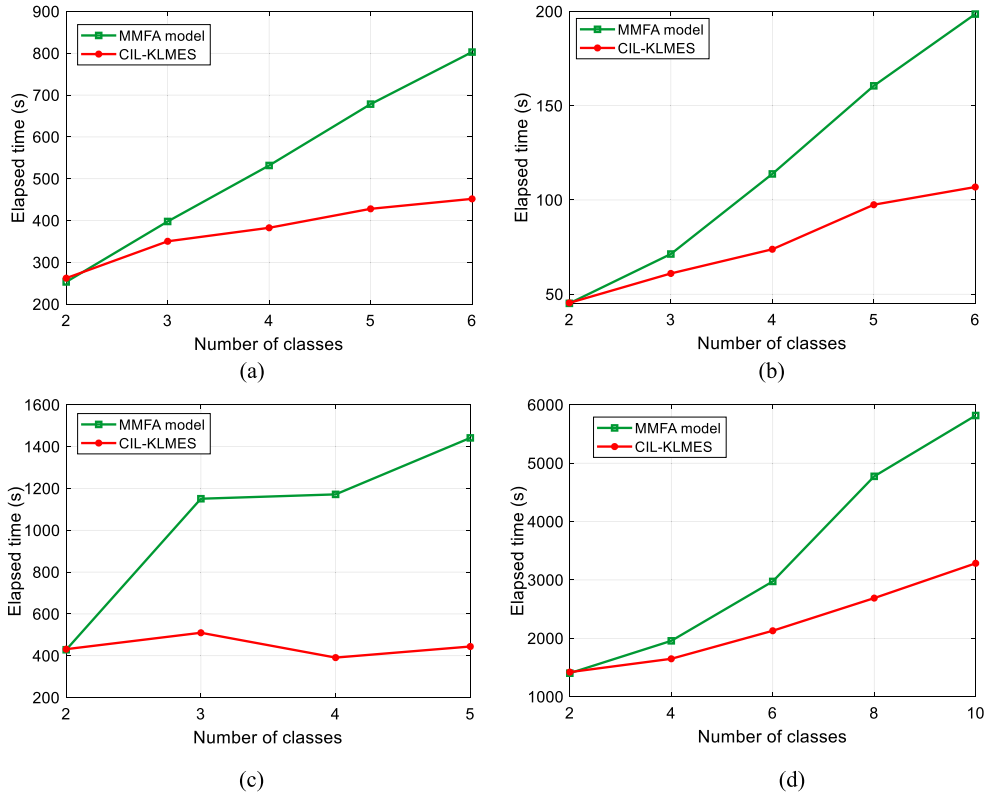


Fig. 5. The variation of time consumption with the arrival of new classes on the 4 datasets: (a) HARUS dataset, (b) GSA dataset, (c) simulated satellite HRRP dataset, and (d) Split CIFAR-10.

shown in Fig. 5. For one thing, our method can converge faster, for another thing, our method utilized less data to update the model, leading to less time consumption.

5.3. Model analysis

5.3.1. Analysis on KL constraint term

Due to the CF problem, the distribution characteristics of old classes will be lost, leading to the confusion between old classes and misclassifying old classes into new classes. In our CIL-KLMES, a KL constraint term is a key factor to alleviate the CF problem. In order to verify the effectiveness of the KL constraint term, we make an analysis on the proposed method with or without KL constraint term in this subsection. In the following, a CIL-KLMES without KL constraint term (i.e., $u = 0$) is referred to as CIL-MES. Note that the exemplar set size of CIL-MES is consistent with our CIL-KLMES. We first take simulated satellite HRRP dataset as an example and present the t-distribution Stochastic Neighbor Embedding (t-SNE) [50] visualizations of features of initial model learned at the initial phase and features of CIL-KLMES and CIL-MES learned at the 1st CIL phase in Fig. 6. In addition, the ratio of between-class distance to within-class distance (BWR) [37] is employed to quantitatively illustrate the effectiveness of our method. The BWR is defined as follows:

$$\begin{aligned} \text{BWR} &= L_{\text{between}} / L_{\text{within}} \\ L_{\text{within}} &= \sum_{z=1}^Z \frac{1}{n_z} \sum_k^{n_z} \left(f_k^{(z)} - \bar{f}^{(z)} \right)^T \left(f_k^{(z)} - \bar{f}^{(z)} \right) \\ L_{\text{between}} &= \sum_{z=1}^Z \left(\bar{f} - \bar{f}^{(z)} \right)^T \left(\bar{f} - \bar{f}^{(z)} \right) \end{aligned} \quad (54)$$

where L_{between} and L_{within} denote the between-class and the within-class distances in the feature space, respectively. Z is the number of classes, and n_z denotes the number of samples of the z th class; $f_k^{(z)}$ is the feature of the k th sample of the z th class. $\bar{f}^{(z)}$ and $\bar{f}$ are the mean feature of the z th class and of all classes, respectively. The larger BWR value indicates that the feature is more separable. As seen in Fig. 6, the t-SNE results present that the features learned by our CIL-KLMES have better inter-class separability and have less confusion between classes than CIL-MES at the 1st CIL phase.

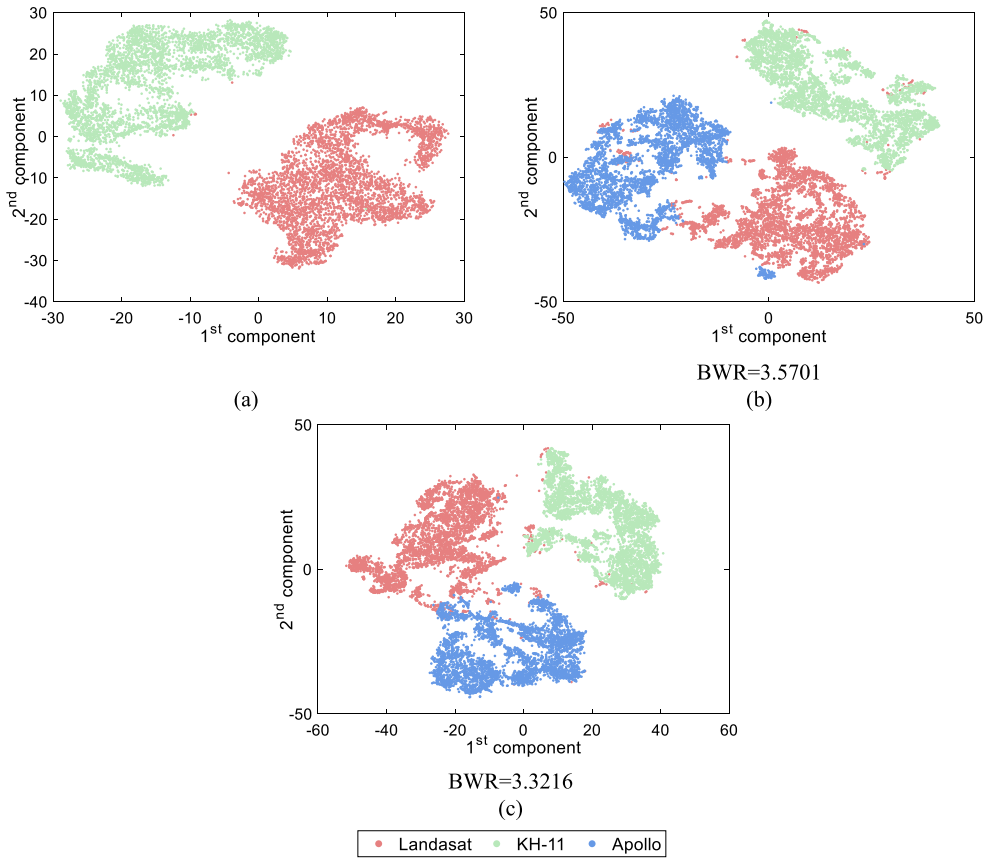


Fig. 6. The t-SNE visualizations of the initial model at the initial phase and CIL-KLMES and CIL-MES at the 1st CIL phase on simulated satellite HRRP dataset. (a) Visualization of the initial model. (b) Visualization of CIL-KLMES. (c) Visualization of CIL-MES.

Then the classification accuracies of CIL-KLMES and CIL-MES are shown in Fig. 7. As shown in Fig. 7, our CIL-KLMES achieves better classification accuracy than CIL-MES, and the classification accuracy degrades more slowly with the increase of new classes. These results demonstrate that adding KL constraint term can maintain the ability of our CIL-KLMES to describe old class data and guarantee the classification performance in dynamic environments.

5.3.2. Analysis on exemplar selection method

Besides the KL constraint term, the exemplar selection method is another crucial factor for alleviating CF problem in our CIL-KLMES. To verify the validity of the proposed exemplar selection method, we compare our CIL-KLMES with CIL-KL, which alleviates the CF problem by only adding the KL constraint term. In addition, we also compare our CIL-KLMES with five CIL methods based on different exemplar selection methods. Herding and Random [28] are two common exemplar selection methods. Replacing our exemplar selection strategy with these two exemplar selection methods, CIL-herdES and CIL-randES are introduced for comparison respectively. Applying density-based exemplar selection strategy in our CIL method (CIL-denES) is also introduced for comparison. Besides these, since our exemplar selection method selects the representative exemplars based on two strategies, i.e., selecting the exemplars with larger likelihood values and selecting the exemplars with closer distance to the classification decision boundary, we respectively select the exemplars through only one strategy as the comparison methods, which are referred to as CIL-ddES and CIL-dbES respectively. For the sake of fairness, the exemplar set size of the above CIL methods, including CIL-herdES, CIL-randES, CIL-denES, CIL-ddES and CIL-dbES, is consistent with our CIL-KLMES.

We first present the t-SNE results taking simulated satellite HRRP dataset as an example to visualize the representative exemplars selected by the above 6 memory-replay-based CIL methods (CIL-KLMES, CIL-herdES, CIL-dbES, CIL-ddES, CIL-randES, CIL-denES) in Fig. 8, which can help us understand the distribution region of the selected exemplars. Specifically, we visualize the features of all 5 classes learned at the 3rd CIL phase (Landsat7, KH-11, Apollo and DongFangHong are old classes, and BeiDou is a new class). In each colored class, light-color points and deep-color ones are the features of the training data and the selected exemplars respectively. In Fig. 8(b), the black stars are the mean of features of different classes. We also compute the BWR values to quantitatively illustrate the effectiveness of our method. From Fig. 8, we can observe that the separability between classes is better through our exemplar selection method. These phenomena mean our exemplar selection method is more beneficial to alleviate the CF problem.

The classification accuracies of 6 memory-replay-based CIL methods and CIL-KL are shown in Fig. 9. It is obvious that our CIL-KLMES shows better accuracies. CIL-randES selects exemplars evenly distributed in each class due to randomness, and the exemplars

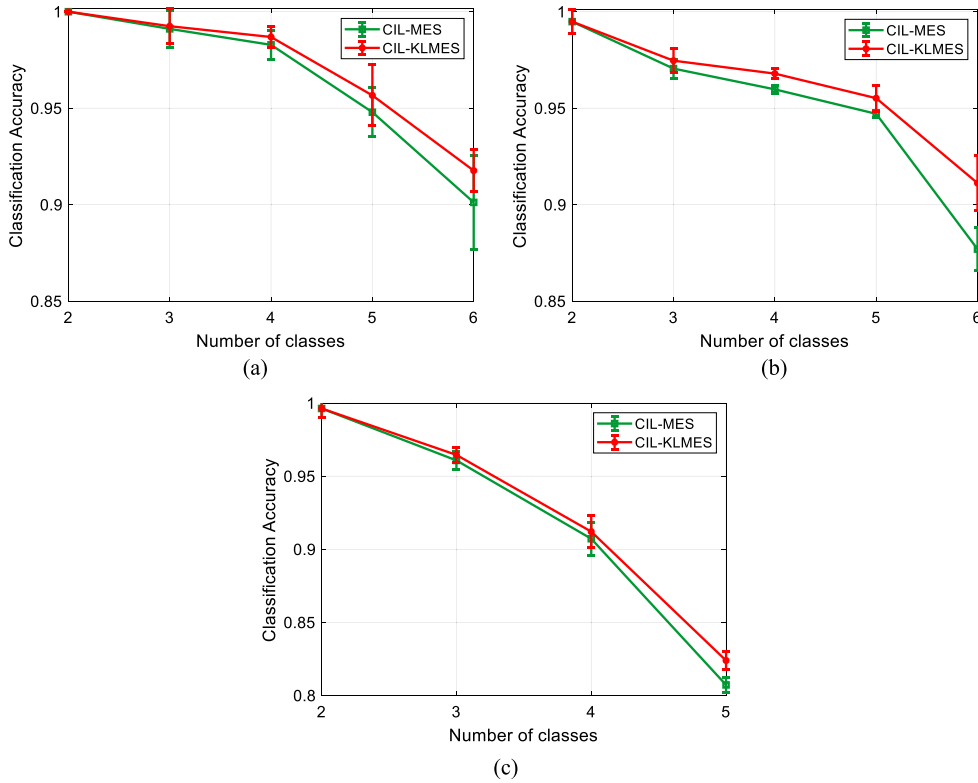


Fig. 7. The variation of classification accuracies of CIL-KLMES and CIL-MES with the arrival of new classes on the 3 datasets: (a) HARUS dataset, (b) GSA dataset, and (c) simulated satellite HRRP dataset.

may scatter in all regions. Nevertheless, some exemplars which are almost unrepresentative may be selected. For the same size exemplar set, the representative exemplars of CIL-randES are relatively few compared with our CIL-KLMES; thus resulting in lower accuracy than our CIL-KLMES. CIL-herdES only considers the center, by which most exemplars are selected near the center of each class. The decision boundary of the old classes cannot be maintained in later CIL phases, which causes the degradation of the separability among all seen classes in the updated feature space. Analogously, the exemplars selected by CIL-denES mainly distributed in areas with higher density, which cannot guarantee maintaining the decision boundary of the old classes. As for CIL-ddES, the representativeness of the exemplars is poor for lack of the information about the class boundary. Most exemplars selected by CIL-dbES concentrate at the class boundary of each class, which may not be helpful to describe the data distribution of old classes in later CIL phases due to the lack of the information about data distribution description, thus weakening classification accuracy. To sum up, our exemplar selection method further ensures the integrality of the knowledge preservation of the past classes, thus better alleviating CF problem at the data level.

5.3.3. Analysis on adding multiple new classes

In real scenarios, multiple new classes may appear at once, i.e., $C_t > 1$. In the following, we investigate the effects of adding multiple new classes at once. First, taking CIFAR-10 datasets as an example, we present the performance of our method and other CIL methods under $C_t = 4$, and the results are shown in Fig. 10. From Fig. 10, we can see that our method achieves better performance when adding more new classes.

Moreover, we analyze the performance of our CIL-KLMES method when adding multiple new classes at once. We take CIFAR-10, HARUS and GSA datasets as examples. We set $C_t = 2$ and $C_t = 4$ for CIFAR-10 dataset. As for HARUS and GSA datasets, we set $C_t = 1$, $C_t = 2$ and $C_t = 4$ respectively. Fig. 11 presents the classification accuracies of our CIL-KLMES under different C_t . With the same number of all seen classes, C_t is inversely proportional to the number of incremental phases. From Fig. 11, we can see that the more classes added at once, the better the classification performance. The main reason is that each update of the model will cause the forgetting, so the more the updates, the more knowledge will be forgotten, leading to lower classification accuracies. That is to say, the forgetting speed of our CIL-KLMES gradually slows down with the increase of C_t , achieving better classification performance.

5.3.4. Analysis on other parameters

It is crucial to analyze the effect of different selection ratios κ . Fig. 12 presents the classification accuracies of our CIL-KLMES on HARUS and GSA datasets under different selection ratios κ : $\kappa = 0.05$, $\kappa = 0.1$ and $\kappa = 0.2$. For the convenience of comparison, the classification accuracies of MMFA model are shown in Fig. 12. We could observe that our CIL-KLMES achieves better performance

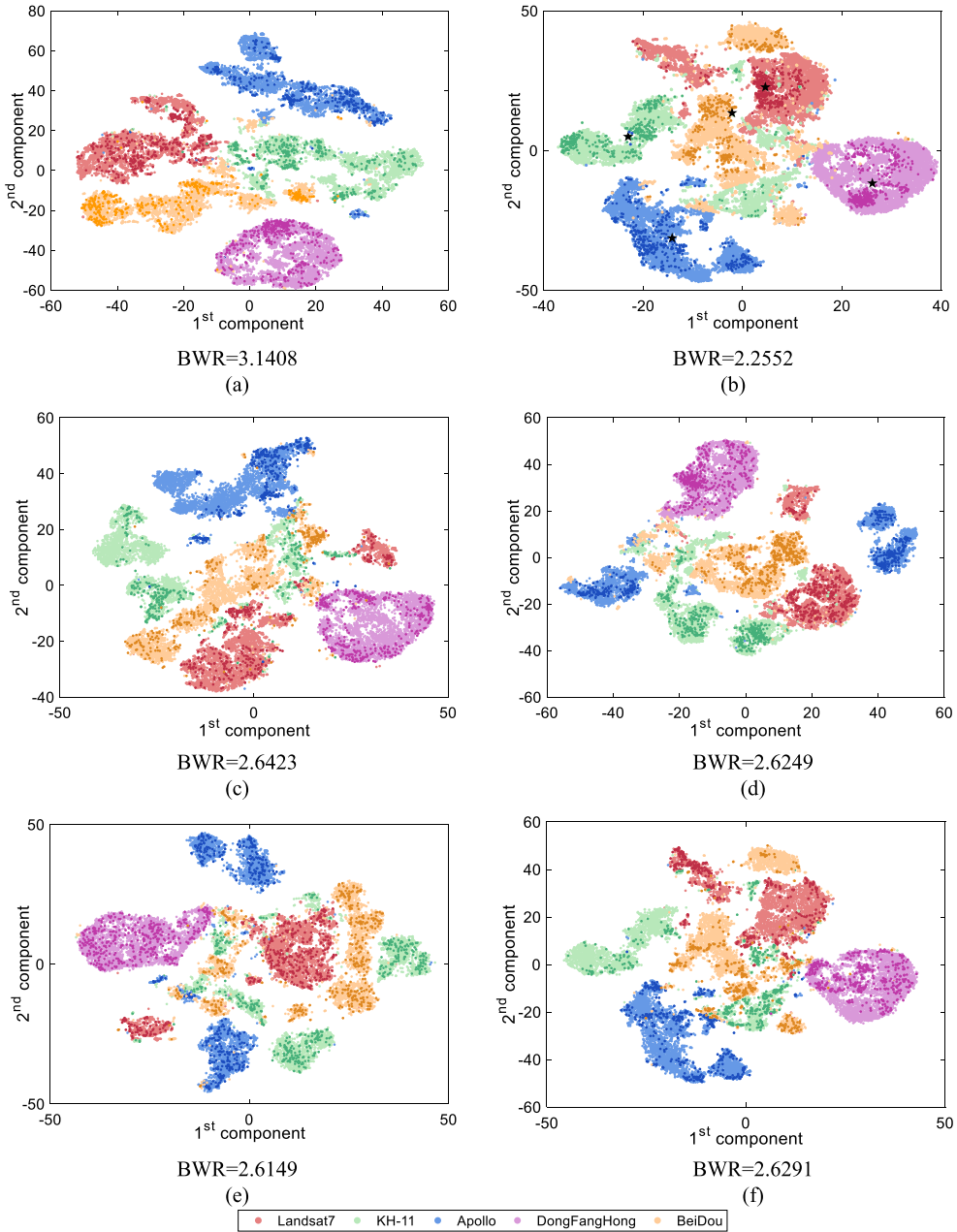


Fig. 8. The t-SNE visualizations of CIL-KLMES based on 6 different exemplar selection methods at the 3rd CIL phase on simulated satellite HRRP dataset. (a) Visualizations of CIL-KLMES. (b) Visualizations of CIL-herdES. (c) Visualizations of CIL-dbES. (d) Visualizations of CIL-ddES. (e) Visualizations of CIL-randES. (f) Visualizations of CIL-denES.

with the increase of selection ratios κ . The main reason is that the more data of old classes are replayed, the less knowledge of old classes is forgotten. However, with the increase of selection ratios κ , the memory requirement becomes large. Thus, the selection ratio κ needs to be set according to the actual situation.

6. Conclusion

We develop a novel CIL-KLMES for classification based on MMFA model. For alleviating the CF problem, CIL-KLMES imposes a KL divergence term to preserve the previous learned knowledge of the past classes at the parameter level. Moreover, a more effective multi-strategy exemplar selection method is developed to select a few representative replay exemplars from two aspects: 1) selecting a few samples with larger likelihood values; 2) selecting a few samples which are close to the classification decision boundary. Replaying these samples helps robustly describe the data distribution and the classification decision boundary of old

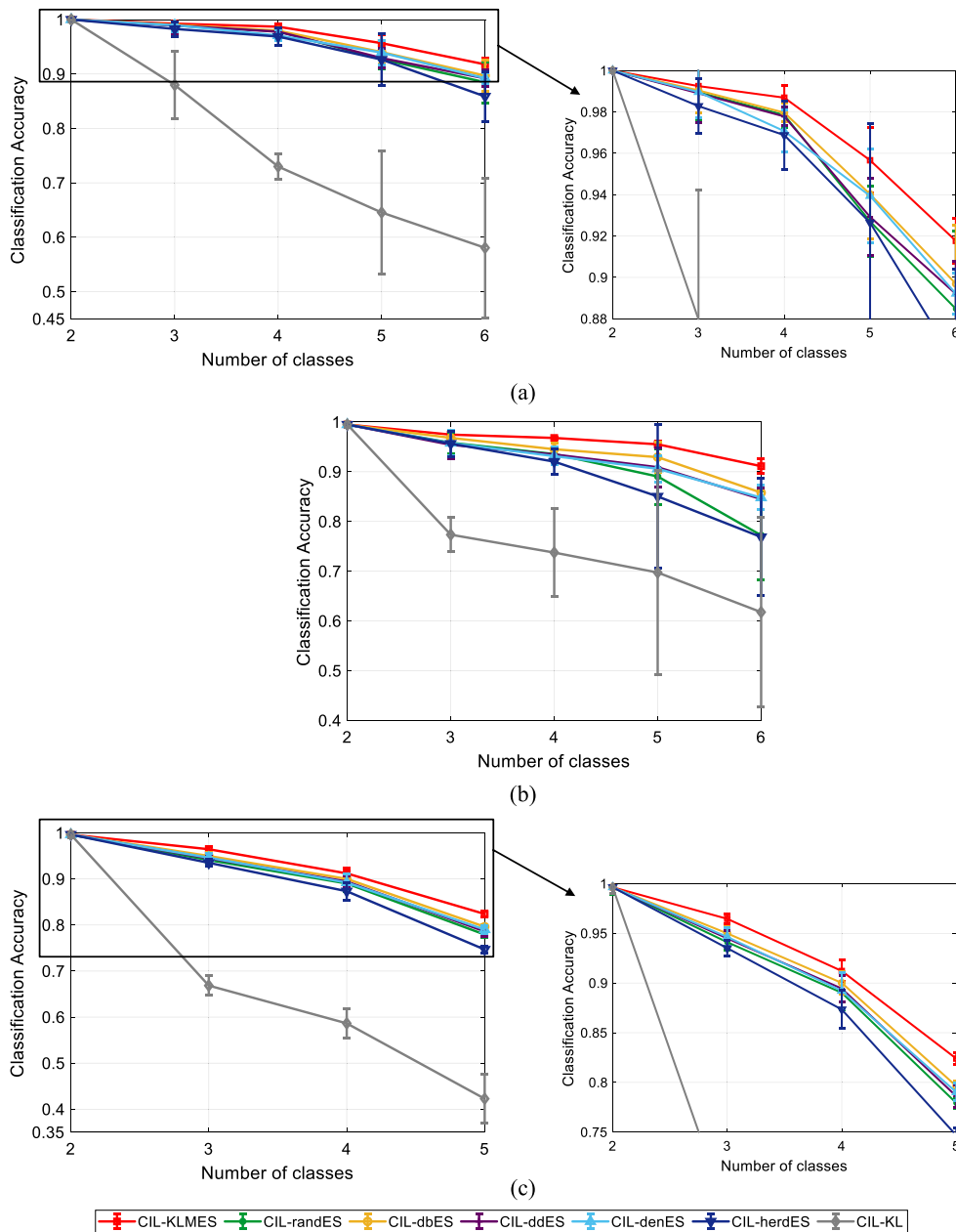


Fig. 9. The variation of classification accuracies of CIL-KLMES, CIL-dbES, CIL-ddES, CIL-herdES, CIL-randES, CIL-denES and CIL-KL with the arrival of new classes on the 3 datasets: (a) HARUS dataset, (b) GSA dataset, and (c) simulated satellite HRRP dataset.

classes. Combining multiple exemplar selection strategies can ensure the integrity of the key knowledge of the past classes, so that the previous learned knowledge of the past classes can be preserved well at the data level. Therefore, the CF problem can be alleviated sufficiently by introducing KL divergence term and replaying the exemplars with more representativeness.

CRedit authorship contribution statement

Yang Li: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Lan Du:** Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jian Chen:** Writing – review & editing, Validation, Methodology, Formal analysis.

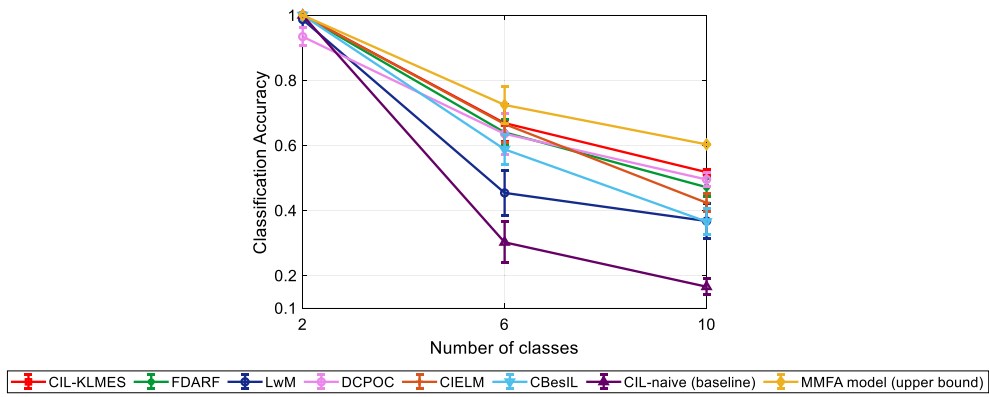


Fig. 10. The classification accuracies of our CIL-KLMES and other CIL methods on CIFAR-10 datasets under $C_t = 4$.

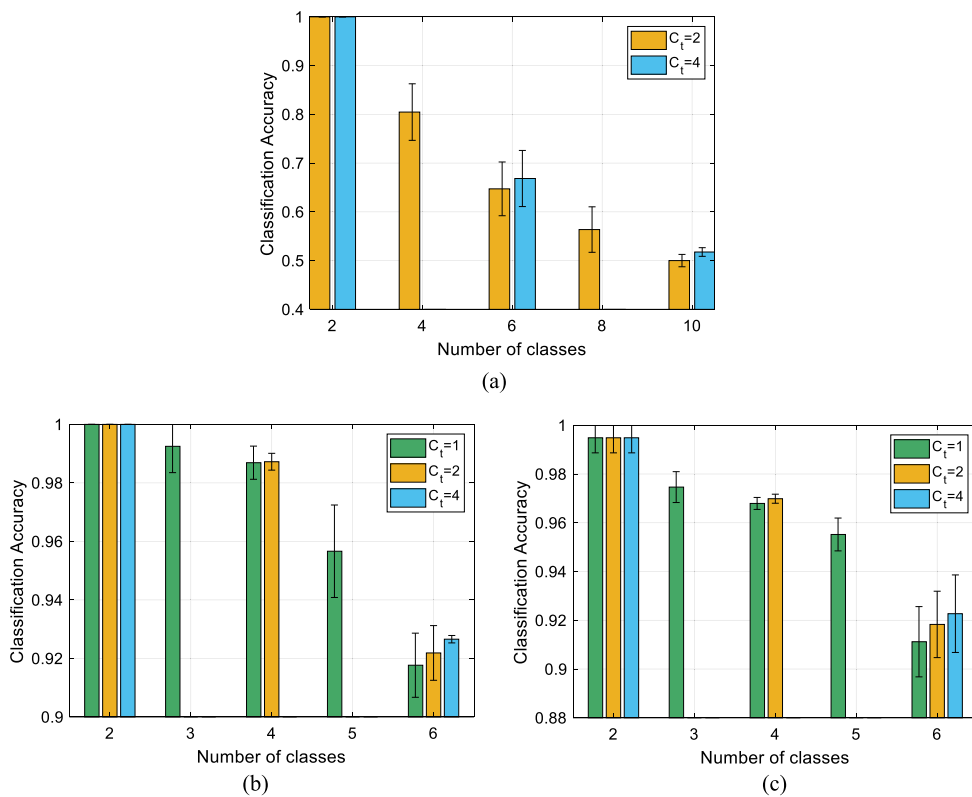


Fig. 11. The classification accuracies of our CIL-KLMES under different C_t on CIFAR-10, HARUS and GSA datasets: (a) CIFAR-10 dataset, (b) HARUS dataset and (c) GSA dataset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

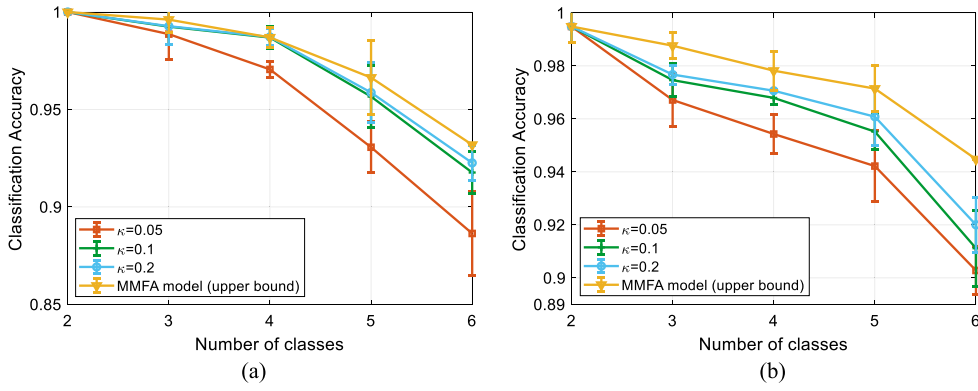


Fig. 12. The classification accuracies of our CIL-KLMES under different selection ratios κ on HARUS and GSA datasets: (a) HARUS dataset, (b) GSA dataset.

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Appendix A

For ease of reading and understanding, Table A.1 below gives the notations of the important variables used in the full text.

Table A.1

List of notations in our CIL-KLMES.

Notation	Meaning
$\hat{\mathbf{X}}^t$	The training set obtained at time t (new class data)
$\mathbf{X}_c^t$	The data of class c in $\hat{\mathbf{X}}^t$
$\mathbf{B}^t$	The exemplar set obtained at time t
$\mathbf{XB}^t$	The training set used at time t with $\mathbf{XB}^t = \{\hat{\mathbf{X}}^t, \mathbf{B}^{t-1}\}$
C_t	The number of classes in $\hat{\mathbf{X}}^t$
C^t	The number of all seen classes at time t
$\mathbf{x}_i^t$	The i th sample in $\mathbf{XB}^t$
y_i^t	The label of $\mathbf{x}_i^t$ with $y_i^t \in \{-1, +1\}$
$\boldsymbol{\mu}^{0:t}$	The mean vector of all seen classes at time t
$\mathbf{D}^t$	The factor loading matrix at time t with $\mathbf{D}^t = [\mathbf{d}_1^t, \dots, \mathbf{d}_k^t, \dots, \mathbf{d}_K^t]$
$\mathbf{s}_i^t$	The latent variable of $\mathbf{x}_i^t$ with $\mathbf{s}_i^t = [s_{i1}^t, \dots, s_{ki}^t, \dots, s_{Kt}^t]$
$\boldsymbol{\beta}_k^t$	The precisions of $\mathbf{s}_i^t$
c_k^{t-1}, d_k^{t-1}	The hyper-parameters of the posterior distribution of $\boldsymbol{\beta}_k^{t-1}$
ϵ_i^t	The noise variable at time t
$\boldsymbol{\gamma}^t$	The precisions of $\boldsymbol{\gamma}_i^t$ at time t with $\text{diag}(\boldsymbol{\gamma}^t) = [\gamma_1^t, \dots, \gamma_p^t, \dots, \gamma_p^t]$
e_p^{t-1}, f_p^{t-1}	The precisions of the posterior distribution of $\boldsymbol{\gamma}_p^{t-1}$
$\mathbf{W}^t$	The weighted coefficient at time t with $\mathbf{W}^t = [\mathbf{w}_0^t, \dots, \mathbf{w}_c^t, \dots, \mathbf{w}_{C^t}^t]$
$\sigma_{\mathbf{w}_c^t}$	The precisions of $\mathbf{w}_c^t$ (if c is a new class)
$\boldsymbol{\tau}_1, \boldsymbol{\tau}_2$	The hyper-parameters of the posterior distribution of $\sigma_{\mathbf{w}_c^t}$
$\tilde{\mathbf{s}}_i^t$	The augmented feature vector at time with $\tilde{\mathbf{s}}_i^t = [\mathbf{s}_i^t; 1]$
λ_i^t	The augmented feature vector at time t
$\boldsymbol{\xi}^t$	The local variables at time t
$\boldsymbol{\theta}^t$	The global variables at time t
h_0	The positive tuning parameter
R	The capacity of the exemplar set
$\mathbf{B}_{dd}^t$	The exemplar set selected based on data distribution description at time t
$\mathbf{B}_{db}^t$	The exemplar set selected based on classification decision boundary preservation at time t
N_{dd}^t	The number of the exemplars in $\mathbf{B}_{dd}^t$
N_{db}^t	The number of the exemplars in $\mathbf{B}_{db}^t$
n_{dd}^c	The number of the exemplars in $\mathbf{B}_{dd}^t$ per class
n_{db}^c	The number of the exemplars in $\mathbf{B}_{db}^t$ per class
P	The dimension of $\mathbf{x}_i^t$
K	The number of factors in $\mathbf{D}^t$
$\mathcal{N}(\cdot)$	The Gaussian distribution
$Ga(\cdot)$	The Gamma distribution

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